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PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Atomic Structure

(English Medium)

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ATOMIC STRUCTURE

1. INTRODUCTION :

The existence of atoms has been proposed since the time of early Indian and Greek philosophers (400 B.C.) who were of the view that atoms are the fundamental building blocks of matter. According to them, the continued subdivisions of matter would ultimately yield atoms which would not be further divisible. The word 'atom' has been derived from the Greek word 'a-tomio' which means 'uncuttable' or 'non-divisible'. These earlier ideas were mere speculations and there was no way to test them experimentally. These ideas remained dormant for a very long time and were revived again by scientists in the nineteenth century. The atomic theory of matter was first proposed on a firm scientific basis by **John Dalton**, a British school teacher in 1808. His theory, called **Dalton's atomic theory**, regarded the atom as the ultimate particle of matter.

2. DALTON ATOMIC THEORY :

(i) Atom is the smallest particle of any substance which cannot be divided further.
(ii) Atoms can never be created nor be destroyed. Only the rearrangement of atoms occurs in chemical reactions.
(iii) All the atoms are hard and dense.
(iv) All the atoms of an element are identical but the atoms of different elements will be different.
(v) A compound is formed by the combinations of atoms of different elements in a fixed ratio by mass.

We start with the experimental observations made by scientists towards the end of nineteenth and beginning of twentieth century. These established that atoms can be further divided into subatomic particles, i.e., electrons, protons and neutrons. a concept very different from that of Dalton. The major problems before the scientists at that time were:

- to account for the stability of atom after the discovery of sub-atomic particles,
- to compare the behaviour of one element from other in terms of both physical and chemical properties,
- to explain the formation of different kinds of molecules by the combination of different atoms and,
- to understand the origin and nature of the characteristics of electromagnetic radiation absorbed or emitted by atoms.

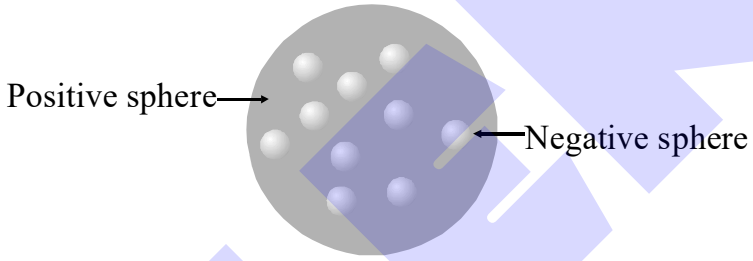
3. ATOMIC MODELS :

Observations obtained from the experiments mentioned in the previous sections have suggested that Dalton's indivisible atom is composed of sub-atomic particles carrying positive and negative charges. Different atomic models were proposed to explain the distributions of these charged particles in an atom. Although some of these models were not able to explain the stability of atoms, two of these models, proposed by J. J. Thomson and Ernest Rutherford are discussed below.

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3.1 *Thomson Model of Atom :*

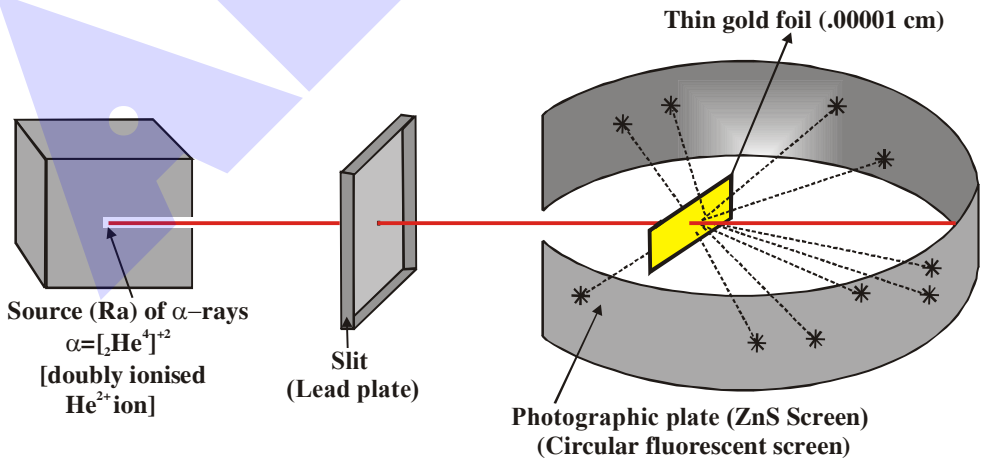
J. J. Thomson, in 1898, proposed that an atom possesses a spherical shape (radius approximately 10^{10} m) in which the positive charge is uniformly distributed. The electrons are embedded into it in such a manner as to give the most stable electrostatic arrangement. Many different names are given to this model, for example, **plum pudding, raisin pudding or watermelon**. This model can be visualised as a pudding or watermelon of positive charge with plums or seeds (electrons) embedded into it. An important feature of this model is that the mass of the atom is assumed to be uniformly distributed over the atom. Although this model was able to explain the overall neutrality of the atom, but was not consistent with the results of later experiments. Thomson was awarded Nobel Prize for physics in 1906, for his theoretical and experimental investigations on the conduction of electricity by gases.



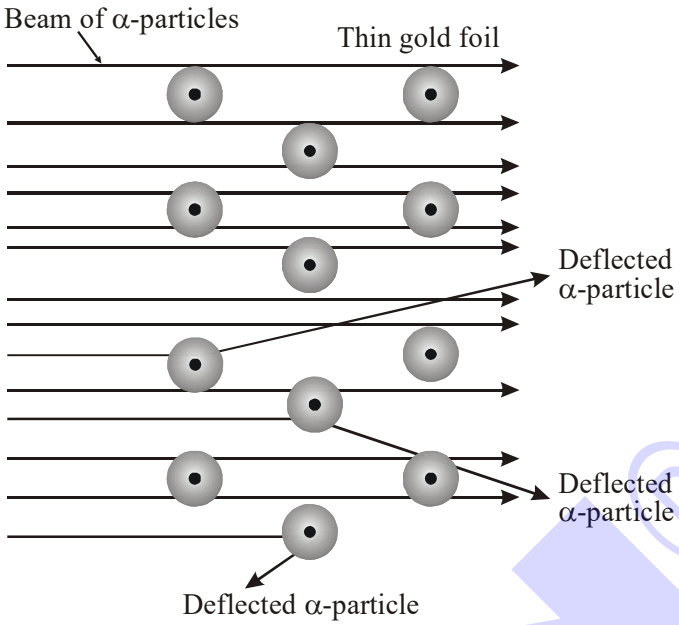
Thomson model of atom

3.2 Rutherford's model :

Rutherford's α-scattering experiment



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B. Schematic molecular view of the gold foil

❖ **Observations and conclusions :**

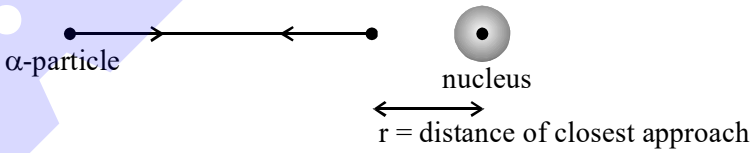
(i) Most of the α-particles passed through the gold foil undeflected. Hence, most of the space in the atom is empty.

(ii) A small fraction of α-particles was deflected by small angles. A very few α-particles (~1 in 20,000) bounced back, that is, were deflected by nearly 180°.

The deflection must be due to enormous repulsive force showing that the positive charge of the atom is not spread throughout the atom as Thomson had presumed. The positive charge has to be concentrated in a very small volume that repelled and deflected the positively charged α-particles. Rutherford given the name **nucleus** to this positively charged center of atom.

(iii) The volume occupied by the nucleus is negligibly small as compared to the total volume of atom. The radius of the atom is about 10^{-10} m, while that of nucleus is 10^{-15} m.

Rutherford estimated the size of nucleus by calculating the **distance of closest approach**.



The initial kinetic energy of α-particle must be equal to potential energy at distance of closest approach.

$$\text{K.E.} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q_1\cdot q_2}{r} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{(2e)\cdot(Ze)}{r}$$

$$\boxed{r = \frac{2Ze^2}{(4\pi\varepsilon_0)(\text{K.E.})}}$$

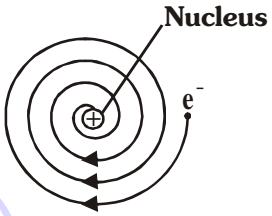
where Z = atomic number of element used in foil.

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(iv) Almost all mass of the atom is densely concentrated in extremely small region (nucleus).

(v) The nucleus is surrounded by electrons that move around the nucleus with a very high speed in circular paths called **orbits**. Electrons and the nucleus are held together by electrostatic forces of attraction.

3.2.1 Drawbacks of Rutherford model :



(i) According to the electromagnetic theory of Maxwell, charged particles when accelerated should emit electromagnetic radiation Therefore, an electron in an orbit will emit radiation, the energy carried by radiation comes from electronic motion. The orbit will thus continue to shrink.

Calculations show that it should take an electron only 10^{-8} s to spiral into the nucleus. But this does not happen. Thus, the Rutherford model cannot explain the stability of an atom.

(ii) It does not explain the arrangement of electrons revolving round nucleus.

(iii) It does not explain the stability of nucleus against strong repulsive forces.

(iv) It does not explain atomic spectrum.

Ex.1 ***An α-particles of kinetic energy of 5.4 MeV is projected towards gold nucleus. Calculate the distance of closet approach. (Atomic number of gold = 79, 1 eV = 1.6 × 10^{-19} J)***

Sol : $\text{K.E.} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q_1q_2}{r}$

or $5.4 \times 10^6 \times 1.6 \times 10^{-19} = 9 \times 10^9 \times \frac{(2\times1.6\times10^{-19})\times(79\times1.6\times10^{-19})}{r}$

$\therefore$ $r = 4.21 \times 10^{-14}$ m

Ex.2 ***An α-particles is projected towards the following nucleus with same kinetic energy in different experiement the distance of closet approch is maximum for***

(A) Na (Z = 11) **(B) Ca (Z = 20)**

(C) Ag (Z = 47) **(D) Au (Z = 79)**

Answer.(A)

Sol. $r \propto Z$

Ex.3 ***An α-particle, a proton, a deutron and a neutron are projected towards the same nucleus with the same kinetic energy in different experiement. The distance of closet approch is minimum for***

(A) α **(B) p** **(C) d** **(D) n**

Answer.(D)

Sol. Neutron is chargeless and hence it will not experience repulsion.

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Ex.4 An α-particle having K.E. = 7.7 MeV is scattered by gold (Z = 79) nucleus through 180°. Find distance of closest approach.

K.E. = 7.7 M eV

$= 7.7 \times 10^6 \times 1.6 \times 10^{-19}$ J

$= 1.23 \times 10^{-12}$ J

$\frac{1}{4\pi\varepsilon_0} = 9\times10^9 \text{Nm}^2\text{C}^{-2}$

Using we get : $\frac{9\times10^9 \times 2\times79\times(1.6\times10^{-19})^2}{1.23\times10^{-12}}$

$r_0 = 3 \times 10^{-14}$ m

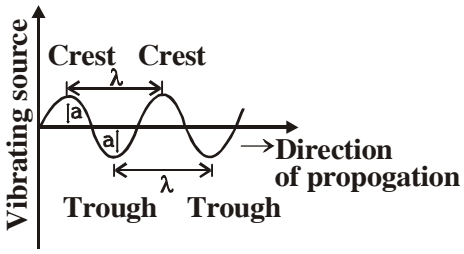
4. *WAVE THEORY :*

A wave is defined as a periodic disturbance in space or in a medium that involves elastic displacement of material particles or a periodic change in some physical quantities such as T, P, V etc. Thus, wave motion represents propagation of a periodic disturbance carrying energy. The wave travels at right angles to the vibratory motion of the object.

When an object moves up and down or vibrates continuously, energy in the form of waves is transmitted by a vibrating object to a distant place. For example , when a stone is thrown in a still water of a pond, a disturbance is produced at a place where the stone strikes the water.

This disturbance advances outwards in the same form and reaches the edges of the pond. Such a disturbance is called a ***mechanical wave***. The mechanical waves transmit only in a material medium. Besides the mechanical waves, there are waves which do not require any medium for their transmission. These waves are called ***electromagnetic waves*** or ***electromagnetic radiations.***

4.1 *WAVE CHARACTERISTICS :*



4.1.1 Wavelength (λ) :

It is defined as the distance between two nearest crests or nearest troughs.

It is measured in terms of a Å (Angstrom), pm (Picometre), nm (nanometer), cm(centimetre), m (metre)

1 Å = 10^{-10} m, 1 pm = 10^{-12} m, 1 nm = 10^{-9} m, 1 cm = 10^{-2} m

4.1.2 Frequency (ν) :

Frequency of a wave is defined as the number of waves which pass through a point in 1 sec.

- It is measured in terms of Hertz (Hz), sec^{-1}, or cycle per second (cps)

  1 Hertz = 1 sec^{-1}

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4.1.3 Time period (T) : Time taken by a wave to pass through one point.

$$T = \frac{1}{\nu} \text{ sec.}$$

4.1.4 Speed (c) :

Speed of a wave is defined as distance covered by a wave in 1 sec.

$c = \lambda / T = \lambda\nu$ $\quad \nu = c/\lambda$ $\quad c = \nu\ (\text{sec}^{-1}) \times \lambda\ (\text{m})$ $\quad c = \nu\lambda\ (\text{m sec}^{-1})$

Since c is constant for em-waves, frequency is inversely propotional to λ

4.1.5 Wave number ($\bar{\nu}$) :

It is defined as number of waves per unit length. It is denoted by $\bar{\nu}$ & is expressed in cm^{-1}.

$$\bar{\nu} = \frac{1}{\lambda} \qquad (1\ cm^{-1} = 100\ m^{-1})$$

- It is measured in terms of cm^{-1}, m^{-1} etc.

4.1.6 Amplitude (a) :

It is the height of the crest or depth of the trough of a wave and is denoted by 'a'. It is half the vertical distance from the top of the wave to the bottom of the wave. It determines the intensity or brightness of the beam of light.

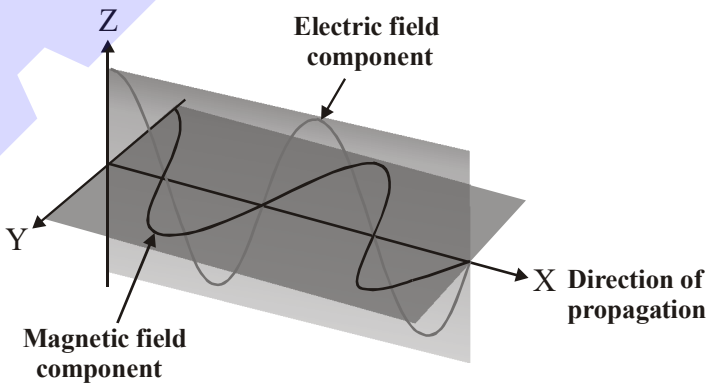
The amplitude of a wave in stretched string is the maximum displacement of the string from its normal position that of water waves is the maximum height of the water surface relative to its normal level, That of a sound wave is the maximum change in pressure relative to the normal pressure.

4.2 ELECTRO MAGNETIC (EM) WAVES :

James Maxwell (1870) suggested that when electrically charged particle moves under acceleration, alternating electrical and magnetic fields are produced and transmitted by it. These fields are transmitted in the forms of waves called **electromagnetic waves** or **electromagnetic radiations**.

4.2.1 Main assumptions of Maxwell EM theory :

(i) The oscillating electric and magnetic fields produced by oscillating charged particles are perpendicular to each other and both are perpendicular to the direction of propagation of the wave.



(The electric and magnetic field components of an electromagnetic wave. These components have the same wavelength , frequency, speed and amplitude, but they vibrate in two mutually perpendicular planes.)

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(ii) Unlike sound waves or water waves, electromagnetic waves do not require medium and can move in vacuum. However speed changes in different mediums

(iii) The EM waves travel with the speed of light (3×10^8 m/sec) in vacuum.

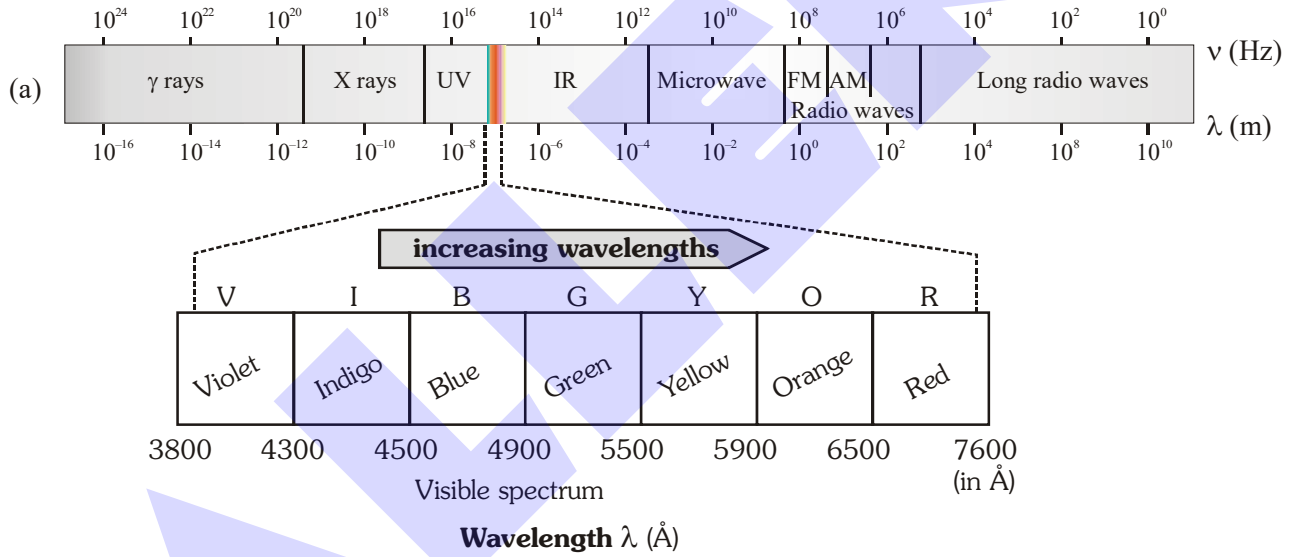
(iv) Energy of EM waves depends on its amplitude not on its frequency or wavelength.

(This assumptions was later proved to be wrong in some situations).

4.3 ELECTRO MAGNETIC SPECTRUM :

The arrangement of various types of electromagnetic radiations in the order of their increasing or decreasing wavelength or frequencies is known as electromagnetic spectrum. The wavelength decreases in the following order.

Radio waves > microwaves > Infrared > Television waves > Visible > Ultraviolet > X-rays > γ-rays > cosmic rays



(a) The spectrum of electromagnetic radiation. (b) Visible spectrum. The visible region is only a small part of the entire spectrum.

Ex.5 *Calculate the frequency of EMR (Electromagnetic radiation) of wave number 10^4 cm^{-1}.*

Sol : $\nu = \frac{c}{\lambda} = c\bar{\nu} = (3 \times 10^8\ m/s)(10^4\ cm^{-1}) = (3 \times 10^{10}\ cm/s)(10^4\ cm^{-1}) = 3 \times 10^{14}\ Hz.$

Ex.6 *A radio station radiate the radiowaves of frequency 20kHz. What is meter band of that radio station?*

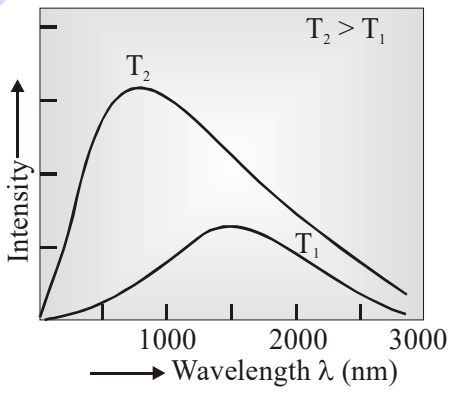
Sol : $20\text{kHz} = \frac{3\times10^8}{\lambda} \Rightarrow \lambda = 1.5 \times 10^4\ m$

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5. PARTICLE NATURE OF ELECTROMAGNETIC RADIATIONS (PLANCK'S QUANTUM THEORY OF RADIATION)

Hot objects emit electromagnetic radiations over a wide range of wavelengths. At high temperatures, an appreciable proportion of radiation is in the visible region of the spectrum. As the temperature is raised, a higher proportion of short wavelength (blue light) is generated. For example, when an iron rod is heated in a furnace, it first turns to dull red and then progressively becomes more and more red as the temperature increases. As this is heated further, the radiation emitted becomes white and then becomes blue as the temperature becomes very high. This means that red radiation is most intense at a particular temperature and the blue radiation is more intense at another temperature. This means intensities of radiations of different wavelengths emitted by hot body depend upon its temperature.

By late 1850's it was known that objects made of different material and kept at different temperatures emit different amount of radiation. Also, when the surface of an object is irradiated with light (electromagnetic radiation), a part of radiant energy is generally reflected as such, a part is absorbed and a part of it is transmitted. The reason for incomplete absorption is that ordinary objects are as a rule imperfect absorbers of radiation. **An ideal body, which emits and absorbs radiations of all frequencies uniformly, is called a black body and the radiation emitted by such a body is called black body radiation.** In practice, no such body exists. Carbon black approximates fairly closely to black body. A good physical approximation to a black body is a cavity with a tiny hole, which has no other opening. Any ray entering the hole will be reflected by the cavity walls and will be eventually absorbed by the walls. A black body is also a perfect radiator of radiant energy. Furthermore, a black body is in thermal equilibrium with its surroundings. It radiates same amount of energy per unit area as it absorbs from its surrounding in any given time. The amount of light emitted (intensity of radiation) from a black body and its spectral distribution depends only on its temperature. At a given temperature, intensity of radiation emitted increases with the increase of wavelength, reaches a maximum value at a given wavelength and then starts decreasing with further increase of wavelength. Also, as the temperature increases, maxima of the curve shifts to short wavelength. Several attempts were made to predict the intensity of radiation as a function of wavelength.



Wavelength intensity relationship

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But the results of the above experiment could not be explained satisfactorily on the basis of the wave theory of light. Max Planck arrived at a satisfactory relationship by making an assumption that absorption and emmission of radiation arises from oscillator i.e., atoms in the wall of black body. Their frequency of oscillation is changed by interaction with oscilators of electromagnetic radiation. Planck assumed that radiation could be sub-divided into discrete chunks of energy.

Max Planck in 1900, put forward a theory known as **"Planck's Quantum Theory".** It regards electromagnetic radiations made up of particles. The main points of the theory are :

(i) The radiant energy is emitted or absorbed by atoms or molecules discontinuously in the form of small energy packets called **quantum**. In case of light , these energy packets are known as **photons**.

(ii) The energy of each quantum is directly proportional to the frequency of the radiation i.e.

$E \propto \nu$ or $E = h\nu = \frac{hc}{\lambda}$

Here h is a constant known as **Planck's constant.** Its value is 6.626×10^{-34} Joules sec.

(iii) The total amount of energy emitted or absorbed by a body is some whole number multiple of quantum i.e.,

$E = nh\nu$ (Here n is a positive integer : 1, 2, 3, 4 ..... etc.)

Thus, Planck for the first time has given a relationship between the frequency (or wavelength) of the radiations and the energy associated with them. Cosmic rays, gamma rays and X-rays are high energy radiations since they have very high frequency. At the same time, microwaves and radiowaves with small frequency are regarded as low energy radiations.

❖ Important features of Photon :

- Source of energy (light) emits radiation in the form photons, which travel with speed of light.
- Energy of single photon, $E = h\nu = \frac{hc}{\lambda}$.
- All the photons in vacuum travel with speed 3×10^8 m/s but their speed is changed in different medium, however frequency remains same. Speed does not depend on energy.
- Photons travel as waves but are absorbed or emitted as particles.

Ex.7 Calculate the energy per quanta of an EMR of frequency 400 MHz.

Sol : $E = 6.626 \times 10^{-34} \times 4 \times 10^6$ *J/quanta*

Ex.8 Calculate the energy per quanta of an EMR of wavelength 662.6 nm.

Sol : $E = n\frac{hc}{\lambda} = \frac{6.626\times10^{-34}\times3\times10^8}{662.6\times10^{-9}\,m} = 3 \times 10^{-19}$ *J/quanta*

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Ex.9 Calculate the wavelength (in Å) of an EMR of energy 3.1 eV/quanta.

Sol: $E = h\nu = h\dfrac{c}{\lambda} \Rightarrow E(ev) \times 1.602 \times 10^{-19} = \dfrac{1\times 6.62\times 10^{-34}\times 3\times 10^{8}}{\lambda(\text{Å})}$

$$\boxed{E(eV) \approx \frac{12400}{\lambda(\text{Å})} = \frac{1240}{\lambda(nm)}}$$

Ex.10 In order to see an object, 10^{-19} J must be recived by our eyes. How many photons of green light must be recieved by our eyes for its visibility. (λ = 550 nm, h = 6.6×10^{-34} J)

Sol : $E = n\dfrac{hc}{\lambda}$

$$10^{-19} = n\times\frac{6.6\times 10^{-34}\times 3\times 10^{8}}{5500\times 10^{-9}}$$

$$n = \frac{5}{18} \approx 1$$

Ex.11 A bulb is rated as 110 watt. If it emits 25% of absorbed energy as red light (λ = 6626Å), how many photons are emitted out by the bulb per second.

Sol: $110\times\dfrac{25}{100} = n\times\dfrac{6.626\times 10^{-34}\times 3\times 10^{8}}{6626\times 10^{-10}}$

Ex.12 The wavelength of microwave radiation is 0.08 m. How many moles of photons is needed to increase the temperature of 400 gm water from 25° to 45°C, assuming 25% efficiency.

Specific heat capacity = 4.2 J/K-gm

Sol : $\left(\dfrac{nhc}{\lambda}\right)\times\dfrac{25}{100} = ms\Delta t$

$$n\times\frac{6.626\times 10^{-34}\times 3\times 10^{8}}{0.08}\times\frac{25}{100} = 400 \times 4.2 \times 20$$

$\therefore$ *Number of moles of photon* $= \dfrac{n}{N_A}$

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Ex.13 A dye absorbs the radiation of 4000 Å and fluoresces the radiation of 5000 Å. If only 40% of the absorbed energy is emitted out, calculate the ratio of number of quanta emitted out and the number of quanta absorbed.

Sol: $E_a \times \frac{40}{100} = E_e$

$$n_a \times \frac{hc}{4000\text{Å}} \times \frac{40}{100} = n_e \times \frac{hc}{5000\text{Å}}$$

$$\frac{n_e}{n_a} = \frac{40}{100} \times \frac{5000}{4000} = \frac{1}{2}$$

Ex.14 The bond dissociation energy of Cl–Cl bond in chlorine gas is 240 kJ/mol. Calculate the longest wavelength of EMR needed to dissociate bond. Assume one photon may dissociate only one bond.

Sol : $\frac{240\times10^3}{6\times10^{23}} = \frac{1\times6.626\times10^{-34}\times3\times10^8}{\lambda}$

Ex.15 A near ultra violet photon of wavelength 300 nm is absorbed by a gas and then emitted as two photons. One photon is of red light with wavelength 760 nm. What would be the wave length of the second photon ?

Sol. It may noted that energy of photon which adsorbed is emitted as sum of the energy of two photons.

Energy absorbed $h\nu = \frac{hc}{\lambda}$

According to available information,

$$\frac{hc}{\lambda} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}; \frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}; \frac{1}{\lambda_2} = \left[\frac{1}{\lambda} - \frac{1}{\lambda_1}\right]$$

Now, $\lambda = 300$ nm ; $\lambda_1 = 760$ nm ; λ_2 can be calculated as :

$$\frac{1}{\lambda_2} = \left[\frac{1}{300} - \frac{1}{760}\right] = \frac{760-300}{300\times760}(\text{nm}^{-1})$$

$$\frac{1}{\lambda_2} = \frac{460}{760\times300}(\text{nm}^{-1})$$

or $\lambda_2 = 496$ nm

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6. PHOTOELECTRIC EFFECT :

Hertz in ***1887*** observed that when a light of certain frequency strikes the surface of a metal, electrons are ejected from the metal. This phenomenon is known as **photoelectric effect** and the ejected **electrons** are called **photoelectrons**.

A few metals, which are having low ionisation energy like Cesium, show this effect under the action of visible light but many more show it under the action of more energetic ultraviolet light.

6.1 The experimental findings are summarized as below :

(i) Electrons come out as soon as the light (of sufficient energy) strikes the metal surface. There is no time lag between the two events.

(ii) The light of any frequency will not be able to cause ejection of electrons from a metal surface.

For each metal, there is a minimum frequency, called the **threshold (or critical) frequency**, which can just cause the ejection. This frequency varies with the nature of the metal. The higher the frequency of the light, the more energy the photoelectrons have. Blue light results in faster electrons than red light.

(iii) Photoelectric current is increased with increase in intensity of light of same frequency, if emission is permitted, i.e. a bright light yields more photoelectrons than a dim one of the same frequency, but the electron energies remain the same.

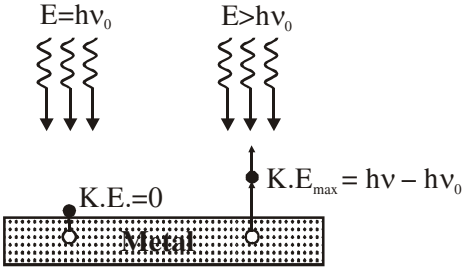
(iv) The number of electrons ejected is proportional to the intensity or brightness of light.

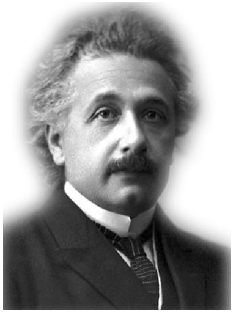
All the above results could not be explained on the basis of laws of classical physics. According to latter, the energy content of the beam of light depends upon the brightness of the light. In other words, number of electrons ejected and kinetic energy associated with them should depend on the brightness of light. It has been observed that though the number of electrons ejected does depend upon the brightness of light, the kinetic energy of the ejected electrons does not. For example, red light [$\nu = (4.3 \text{ to } 4.6) \times 10^{14}$ Hz] of any brightness (intensity) may shine on a piece of potassium metal for hours but no photoelectrons are ejected. But, as soon as even a very weak yellow light ($\nu = 5.1–5.2 \times 10^{14}$ Hz) shines on the potassium metal, the photoelectric effect is observed. The threshold frequency (ν_0) for potassium metal is 5.0×10^{14} Hz. Einstein (1905) was able to explain the photoelectric effect using Planck's quantum theory of electromagnetic radiation as a starting point.

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6.2 Einstein's explanation :

Light must have stream of energy particles or quanta of energy (hν). Suppose, the threshold frequency of light required for ejecting electrons from a metal is ν_0, when a photon of light of this frequency strikes a metal it imparts its entire energy ($h\nu_0$) to the electron.





Albert Einstein (1879 - 1955)

Albert Einstein, a German born American physicist, is regarded by many as one of the two great physicists the world has known (the other is Isaac Newton). His three research papers (on special relativity, Brownian motion and the photoelectric effect) which he published in 1905, **Albert Einstein (1879 - 1955)** while he was employed as a technical assistant in a Swiss patent office in Berne have profoundly influenced the development of physics. He received the Nobel Prize in Physics in 1921 for his explanation of the photoelectric effect.

This energy enables the electron to break away from the surface by overcoming the attractive influence of the nucleus. Thus each photon can eject one electron. If the frequency of light is less than ν_0 , there is no ejection of electron. If the frequency of light is higher than ν_0 (let it be ν), the photon of this light having higher energy (hν), will impart some energy to the electron that is needed to remove it away from the atom. Einstein proposed that light consisted of quanta, which we call photons with a frequency over a certain threshold would have sufficient energy to eject a single electron, producing the photoelectric effect.

6.3 Einstein's Equation for the Photoelectric Effect :

Einstein's interpretation of the photoelectric effect results in equation :

Energy of photon = Energy needed to remove an electron + Max. Kinetic energy of the emitted electron

The excess energy would give a certain velocity (i.e. kinetic energy) to the electron.

$$h\nu = h\nu_0 + K.E._{max.}$$

$$h\nu = h\nu_0 + \frac{1}{2}mv^2_{max}$$

$$\frac{1}{2}\ mv^2_{max} = h\nu - h\nu_0$$

where ν = frequency of the incident light , ν_0 = threshould frequency

v_{max} = max. speed of photoelectron.

$h\nu_0$ is the **threshold energy** (or) the **work function** denoted by $\phi = h\nu_0$ (minimum energy of the photon to liberate electron). It is constant for a particular metal.

The maximum kinetic energy of the photoelectrons increases linearly with the frequency of incident light. Thus, if the energy of the ejected electrons is plotted as a function of frequency, it results in a straight line whose slope is equal to Planck's constant 'h' and whose intercept is $h\nu_0$.

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6.4 Important conclusions from photoelectric effect :

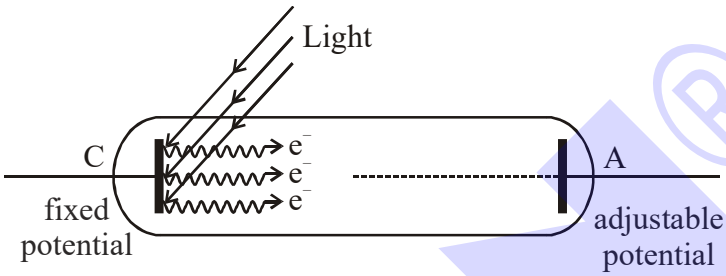
(i) Photoelectric effect demonstrates particle nature of radiation.

(ii) A photon is quanta of energy. Its rest mass is zero. This is why photon can give up its all energy to the particle it strikes.

(iii) There is no effect of frequency of incident light on the number of the emitted photoelectrons.

(iv) There is no effect of intensity of incident light on the K.E. of the emitted photoelectrons.

6.5 SATURATION CURRENT & STOPPING POTENTIAL



Case-I : $V_C = V_A$

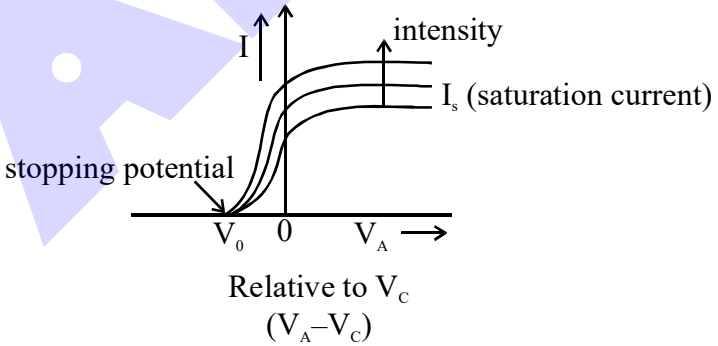
Some of the ejected electrons reach at electrode A resulting **photocurrent**.

Case-II : $V_C < V_A$

As electrode (A) is at high potential, it attracts the electron & even a slower electron will reach at electrode A. It will result in increase in photocurrent. Further increase in the potential difference, a situation may result when the slowest photocurrent electron reach at electrode. It results maximum photocurrent called **saturation current** . Further increase in potential will not increase photocurrent.

Case-III : $V_C > V_A$

As electrode (A) is at low potential, it will repel electron resulting decrease in photocurrent. Further decrease in potential at electrode (A) may result a situation when the fastest electron just fails to reach at (A) and the photocurrent drops to zero. The pontential of (A) relative to C to just stop photocurrent is called **stopping potential**.



On increasing the intensity of light the stopping potential does not change because the maximum K.E. of photoelectron does not change. But the photocurrent increases, because the number of photon falling on the surface increase.

If the frequency of light is changed, the stopping potential will change.

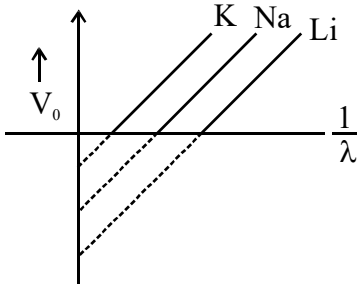
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$(KE)_{max} = eV_0$

$(KE)_{max} = h\nu - h\nu_0$

$eV_0 = \frac{hc}{\lambda} - \phi$

$V_0 = \frac{hc}{e}\frac{1}{\lambda} - \frac{\phi}{e}$



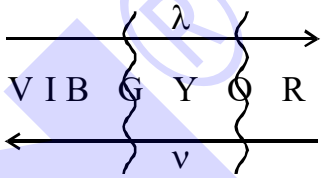
Ex.16 From a metal surface, photoelectron never comes out by orange light but comes from green light. Predict about the injection of photoelectron from the same metal by

(i) Red (ii) Blue (iii) Yellow light

Sol: ***(i)*** *Red light = No*

(ii) *Blue light = Yes*

(iii) *Yellow light = Can't say*



Ex.17 The work function of a metal is 3 eV. If EMR of 200 nm fall on the metal surface, calculate the maximum speed of photoelectron ejected.

Sol: $E = \frac{1240}{200} = 6.2eV$

$(KE)_{max} = h\nu - \phi$

$\frac{1}{2}mv^2 = (6.2-3)eV$

$\frac{1}{2}\times 9.1\times 10^{-31} v_{max}^2 = 3.2\times 1.6\times 10^{-19}$

$\therefore\ v_{max} = 1.06 \times 10^6\ ms^{-1}$

Ex.18 When EMR of frequency 5×10^{15} Hz fall on a metal surface, the maximum kinetic energy of photoelectron is double than the photoelectron which emitts when EMR of frequency 3×10^{15} Hz fall on the same metal. The thresold frequency for the metal is

Sol. $(K.E.)_1 = h \times 5 \times 10^{15} - h\nu_0$

$(K.E.)_2 = h \times 3 \times 10^{15} - h\nu_0$

$2E_1 = E_2$

$\nu_0 = 1 \times 10^{15}$ Hz.

Ex.19 A photon of wavelength 3000 Å strikes a metal surface, the work function of the metal being 2.20 eV. Calculate (i) the energy of the photon in eV (ii) the kinetic energy of the emitted photo electron and (iii) the velocity of the photo electron.

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Sol. (i) Energy of the photon

$$E = h\nu = \frac{hc}{\lambda} = \frac{(6.6\times10^{-34}\,Js)(3\times10^{8}\,ms^{-1})}{3\times10^{-7}\,m} = 6.6 \times 10^{-19}\ J$$

$1\ eV = 1.6 \times 10^{-19}\ J$

$$\text{Therefore } E = \frac{6.6\times10^{-19}\,J}{1.6\times10^{-19}\,J/eV} = 4.125\ eV$$

(ii) Kinetic energy of the emitted photo electron

Work function = 2.20 eV

Therefore, KE = 2.475 – 2.20

= 1.925 eV = 3.08×10^{-19} J

(iii) Velocity of the photo electron

$$KE = \frac{1}{2}mv^2 = 3.08 \times 10^{-19}\ J$$

$$\text{Therefore, velocity (v)} = \sqrt{\frac{2\times3.08\times10^{-19}}{9.1\times10^{-31}}} = 8.22 \times 10^{5}\ ms^{-1}$$

7. SPECTRUM

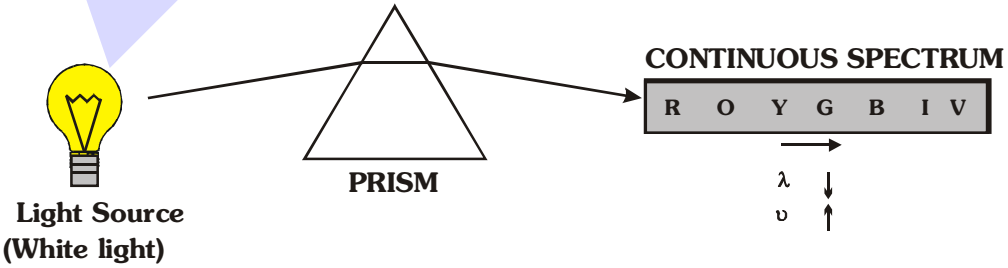
It is the impressions produced on any screen when a light falls on it after passing through prism or prism like material.

Classification of spectrum :

(i) Continuous and discontinuous spectrum :

In continuous spectrum, the impression produced on screen overlap each other, but in discontinous spectrum, some gap exist between the impression.

The spectrum of white light that we can see ranges from violet at 7.50×10^{14} Hz to red at 4×10^{14} Hz. This spectrum is called *continuous spectrum* because violet merges into blue, blue into green, and so on. A similar spectrum is produced when a rainbow forms in the sky. In a continuous spectrum, radiations corresponding to all the wavelengths are present.



(ii) Emission and Absorption Spectra : The spectrum of radiation emitted by a substance that has absorbed energy is called an **emission spectrum**. Atoms, molecules or ions that have absorbed radiation are said to be **'excited'**. To produce an emission spectrum, energy is supplied

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to a sample by heating it or irradiating it and the wavelength (or frequency) of the radiation emitted, as the sample gives up the absorbed energy, is recorded.

An absorption spectrum is like the photographic negative of an emission spectrum. A continuum of radiation is passed through a sample which absorbs radiation of certain wavelengths. The missing wavelength which corresponds to the radiation absorbed by the matter, leave dark spaces in the bright continuous spectrum.

The study of emission or absorption spectra is referred to as **spectroscopy**. The spectrum of the visible light, as discussed above, was continuous as all wavelengths (red to violet) of the visible light are represented in the spectra. The emission spectra of atoms in the gas phase, on the other hand, do not show a continuous spread of wavelength from red to violet, rather they emit light only at specific wavelengths with dark spaces between them. Such spectra are called **line spectra** or **atomic spectra** because the emitted radiation is identified by the appearance of bright lines in the spectra.

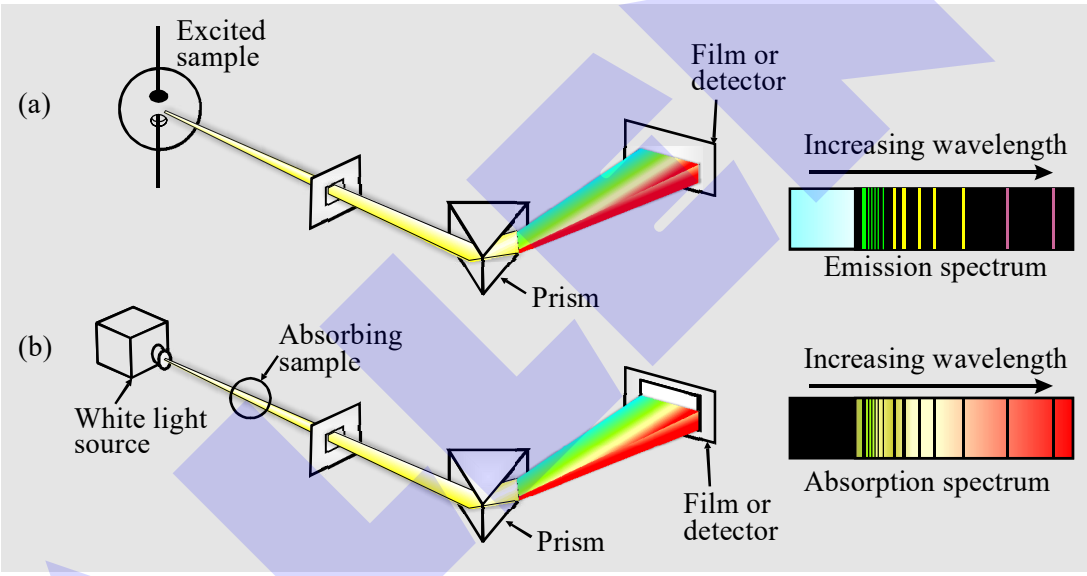


***Fig. (a) Atomic emission**. The light emitted by a sample of excited hydrogen atoms (or any other element) can be passed through a prism and separated into certain discrete wavelengths. Thus an emission spectrum, which is a photographic recording of the separated wavelengths is called as line spectrum. Any sample of reasonable size contains an enormous number of atoms. Although a single atom can be in only one excited state at a time, the collection of atoms contains all possible excited states. The light emitted as these atoms fall to lower energy states is responsible for the spectrum. **(b) Atomic absorption**. When white light is passed through unexcited atomic hydrogen and then through a slit and prism, the transmitted light is lacking in intensity at the same wavelengths as are emitted in (a) The recorded absorption spectrum is also a line spectrum and the photographic negative of the emission spectrum.*

Line emission spectra are of great interest in the study of electronic structure. Each element has a unique line emission spectrum. The characteristic lines in atomic spectra can be used in chemical analysis to identify unknown atoms in the same way as finger prints are used to identify people. The exact matching of lines of the emission spectrum of the atoms of a known element with the lines from an unknown sample quickly establishes the identity of the latter, German

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chemist, Robert Bunsen (1811-1899) was one of the first investigators to use line spectra to identify elements.

Elements like rubidium (Rb), caesium (Cs) thallium (Tl), indium (In), gallium (Ga) and scandium (Sc) were discovered when their minerals were analysed by spectroscopic methods. The element helium (He) was discovered in the sun by spectroscopic method.

8. BOHR'S ATOMIC MODEL

Bohr's model for hydrogen atom is based on the following postulates:



Niels Bohr (1885-1962)

Niels Bohr, was a Danish physicst. After first world war, Bohr worked for peaceful uses of atomic energy. He was awarded the Nobel Prize in physics in 1922.

(i) The electron in the hydrogen atom can move around the nucleus in a circular path of fixed radius and energy. These paths are called **orbits**, **stationary states** or **allowed energy states**. These orbits are arranged concentrically around the nucleus.

(ii) The energy of an electron in the orbit does not change with time. However, the electron will move from a lower stationary state to a higher stationary state when required amount of energy is absorbed by the electron or energy is emitted when electron moves from higher stationary state to lower stationary state. The energy change does not take place in a continuous manner.

(iii) The frequency of radiation absorbed or emitted when transition occurs between two stationary states that differ in energy by ΔE, is given by :

$$\nu = \frac{\Delta E}{h} = \frac{E_2 - E_1}{h}$$

where E_1 and E_2 are the energies of the lower and higher allowed energy states respectively. This expression is commonly known as **Bohr's frequency rule**.

(iv) The angular momentum of an electron in a given stationary state can be expressed as in equation

$$m_e vr = n \cdot \frac{h}{2\pi} \qquad n = 1, 2, 3$$

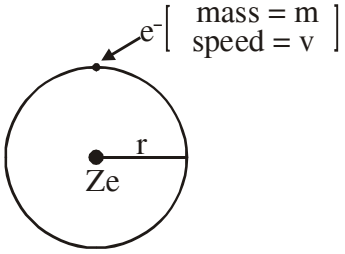
Thus an electron can move only in those orbits for which its angular momentum is integral multiple of $h/2\pi$. That is why, only certain fixed orbits are allowed.

8.1 APPLICATION OF BOHR'S MODEL

When electron revolves in fixed circular orbit than electrostatic force of attraction and centripetal force are equal.

$$\text{Electrostatic force} = \frac{Kq_1q_2}{r^2} = \frac{K.Ze.e}{r^2} = \frac{KZe^2}{r^2}$$

Where, constant $K = 9 \times 10^9\ Nm^2/C^2$ (MKS) = 1 (CGS)



$$\text{Centrifugal force} = \frac{mv^2}{r}$$

In balanced condition

Electrostatic force = Centrifugal force

$$\frac{KZe^2}{r^2} = \frac{mv^2}{r} \text{ or } \frac{KZe^2}{r} = mv^2 \qquad \text{.....(i)}$$

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8.1.1 Radius of various orbits (shell) :

According to Bohr model, $mvr = \frac{nh}{2\pi}$

$$v = \frac{nh}{2\pi mr} \qquad \text{.....(ii)}$$

Now putting the value of v from eq.(ii) into eq.(i)

$$\frac{KZe^2}{r} = m\left(\frac{nh}{2\pi mr}\right)^2$$

$$\frac{KZe^2}{r} = \frac{mn^2h^2}{4\pi^2m^2r^2}$$

$$\boxed{r = \frac{n^2h^2}{4\pi^2mKZe^2}} \qquad \text{.....(iii)}$$

Putting the value of π, h, m, K, & e (Constants) in the above eq .(iii)

$r = 0.529 \times 10^{-10} \times \frac{n^2}{Z}$ m $\qquad \{ Å = 10^{-10}\,m = 10^{-8}\,cm\}$

or $\boxed{r_n = 0.529 \times \frac{n^2}{Z} Å}$

8.1.2. Speed of electron in Bohr orbit :

According to Bohr postulate

$$mvr = \frac{nh}{2\pi}$$

$$v = \frac{nh}{2\pi mr} = \frac{nh}{2\pi m \times n^2h^2 / 4\pi^2 mKZe^2}$$

$$\boxed{v = \frac{2\pi KZe^2}{nh}} \qquad \text{.....(iv)}$$

Putting the value of π, h, K, & e (Constants) in the above eq (iv)

$$\boxed{v = 2.18 \times 10^6 \frac{Z}{n} m/s}$$

8.1.3. Total energy of electron in Bohr orbit :

Total energy of an electron is the sum of kinetic and potential energy.

i.e. T.E. = K.E.+ P.E.

(i) Potential energy : $P.E. = -\frac{Kq_1q_2}{r} = = -\frac{KZe^2}{r} = -\frac{KZe^2}{r}$

(ii) Kinetic energy : $K.E. = \frac{1}{2}mv^2$

But $\frac{KZe^2}{r} = mv^2$ (By eq. i)

$K.E. = \frac{KZe^2}{2r}$

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(iii) Total energy : T.E. = K.E.+ P.E.

$$T.E. = \frac{KZe^2}{2r} - \frac{KZe^2}{r} = -\frac{KZe^2}{2r}$$

Now putting the value of r from eq. (iii)

$$T.E. = -\frac{KZe^2 \times 4\pi^2 mKZe^2}{2n^2h^2}$$

$$\boxed{\mathbf{T.E. = -\frac{2\pi^2 m \times K^2Z^2e^4}{n^2h^2}}}$$

Now putting the value of π, K, e, m, h, we get :

$$T.E. = -2.18\times10^{-18}\times\frac{Z^2}{n^2}\ J/atom = -1312\times\frac{Z^2}{n^2}\ kJ/mol$$

$$= -2.18\times10^{-11}\times\frac{Z^2}{n^2}\ erg/atom = -313.6\times\frac{Z^2}{n^2}\ Kcal/mol$$

$$= -13.6\times\frac{Z^2}{n^2}\ eV/atom$$

8.1.4 Some extra points :

(i) $K.E = \frac{KZe^2}{2r}$ i.e. $K.E. \propto \frac{1}{r}$ On increasing radius, K.E. decreases.

(ii) $P.E. = -\frac{KZe^2}{r}$ i.e. $P.E. \propto \left(-\frac{1}{r}\right)$ On increasing radius, P.E. increases.

(iii) $T.E. = -\frac{KZe^2}{2r}$ i.e. $E. \propto \left(-\frac{1}{r}\right)$ On increasing radius, total energy increases.

(iv) Relation between T.E., P.E. and K.E.

P.E = –2 KE

KE = –T.E.

P.E = 2 T.E.

8.1.5 Important Defination :

(i) Ionization energy :

Minimum energy required to liberate an electron from the ground state of an isolated atom is called the ionization energy.

(ii) Separation energy :

Minimum energy required to remove an electron from its excited state is called as separation energy.

(iii) Excitation energy :

Amount of energy required to shift an electron from ground state to any excited state.

Note : All these kinds of energy are always positive. (We only consider their magnitude)

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Ex.20 Calculate the radius of first four orbits of hydrogen atom

Sol: $r_1 = 0.529 \times \frac{1^2}{1} = 0.529\text{Å}$

$r_2 = 0.529 \times \frac{2^2}{1} = 2.116\text{Å} \ = r_1 \times 2^2$

$r_3 = 0.529 \times \frac{3^2}{1} = 4.761\text{Å} \ = r_1 \times 3^2$

$r_4 = 0.529 \times \frac{4^2}{1} = 8.464\text{Å}$

From this, for same Z : $r_n = r_1 \times n^2$

Ex.21 Calculate the ratio of radius of 2nd orbits of Li^{2+} atom & 3rd orbits He^+ ion.

Sol: $\frac{r_{2,Li^{2+}}}{r_{3,He^+}} = \frac{0.529 \times \frac{4}{3}}{0.529 \times \frac{9}{2}} = \frac{8}{27}$

Ex.22 Calculate the radius ratio of 3rd & 5th orbit of He^+.

Sol. For same z, $r_n = r_1 \, . \, n^2$

$\therefore \frac{r_3}{r_5} = \frac{r_1 \times 3^2}{r_1 \times 5^2} = \frac{9}{25}$

Ex.23 Calculate the energy of Li^{+2} atom for 2nd excited state.

Sol. $E = -13.6 \times \frac{Z^2}{n^2}$

$\because$ Z = 3 and e^- exist in 2nd excited state, means e^- present in 3rd shell i.e. n = 3

$\therefore \quad E = -13.6 \times \frac{(3)^2}{(3)^2} = -13.6$ eV/atom

Ex.24 If the P.E. of an electron is – 6.8 eV in hydrogen atom then find out K.E. of electron and radius of orbit, where electron exist.

Sol. (i) P.E. = – 2K.E.

– 6.8 = – 2K.E.

$\frac{6.8}{2}$ = K.E. $\Rightarrow$ **K.E. = 3.4 eV**

(ii) $\text{T.E.} = \frac{\text{P.E.}}{2} = -3.4$ eV

$\because \quad E = -13.6 \times \frac{Z^2}{n^2}$

or $-3.4 = -13.6 \times \frac{1^2}{n^2}$

$\therefore \quad n = 2$

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(iii) $r = 0.529 \times \frac{n^2}{Z}$ Å

$r = 0.529 \times \frac{(2)^2}{1}$ Å

$= 0.529 \times 4$Å $= 2.16$ Å

Ex.25 Calculate the required energy in eV to excite electron from the ground state to 1st excited state in a hydrogen atom.

Sol. $E_n = -13.6\frac{z^2}{n^2}$ eV

$E_1 = -13.6 \times \frac{1^2}{1^2} = -13.6$ eV

$E_2 = -13.6 \times \frac{1^2}{2^2} = -3.4$ eV

$\therefore$ Required energy $= E_2 - E_1$

$= (-3.4) - (13.6) = 10.2$ eV.

Ex.26 Calculate the amount of energy absorbed in the trasition n = 1 to n = 3 in Li^{2+} ion.

Sol. *n_1 orbit* $\rightleftharpoons$ *n_2 orbit*

$\Delta E = E_{n_2} - E_{n_1} = \left(-13.6\frac{z_1^2}{n_2^2}\right) - \left(-13.6\frac{z_2^2}{n_1^2}\right)$

$$\boxed{\Delta E = 13.6\, z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) eV}$$

$\Delta E = 13.6 \times 3^2\left(\frac{1}{1^2} - \frac{1}{3^2}\right) = 108.8$ eV

Ex.27 Calculate the excitation energy of Be^{3+} ion in ground state.

Sol. $\Delta E = 13.6 \times 16\left(\frac{1}{1} - \frac{1}{4}\right) =$ *163.2 eV*

Ex.28 The ionisation energy of He^+ ion is x kJ/mole. Calculate ionisation energy of Li^{2+} ion.

Sol. *For I.E.* $\Rightarrow n = 1 \longrightarrow n = \infty$

$IE = 13.6\, z^2\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right)$

$\therefore$ $\boxed{\text{I.E.} = 13.6\, z^2 \text{ eV}}$ $\Rightarrow$ *I.E.* $\propto z^2$

$\frac{(\text{I.E.})_{Li^{2+}}}{(\text{I.E.})_{He^+}} = \frac{3^2}{2^2} \Rightarrow (\text{I.E.})_{Li^{2+}} = \frac{9}{4}x$ *kJ/mol*

Ex.29 The ionisation energy for a single electron system is 14.4 eV. Calculate the amount of energy released when electron jumps from 3rd orbits to 2nd orbit.

Sol. $\Delta E = (IE)\left(\frac{1}{n_1^2} - \frac{1}{n^2}\right) = 14.4 \times \left(\frac{1}{4} - \frac{1}{9}\right) =$ *2 eV*

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Ex.30 Calculate the speed of an electron in the 3rd orbit of the Li^{2+} ion. Also calculate the number of revolutions per second that it makes around the nucleus.

Sol. Radius of 2nd orbit = $r_1 x \frac{(n)^2}{Z} = 0.529 \times \frac{(3)^2}{3} = 1.587 Å$

Speed of electron in 2nd orbit, $v = 2.18 \times 10^6 \frac{Z}{n}$ m/sec = 2.18×10^6 m/sec

No. of revolutions/sec = $\frac{1}{2\pi r/v} = \frac{v}{2\pi r} = \frac{2.18\times10^6 m/sec}{2\times3.14\times1.587\times10^{-10} m}$

$= 2.187 \times 10^{15}$ rev/sec

8.2 RYDBERG FORMULA

It an electron shows transition from n_2 to n_1 energy level then energy change ΔE will be.

$$\Delta E = En_2 - En_1$$

$$\Delta E = \left[\frac{-2\pi^2 mK^2Z^2e^4}{n_2^2h^2}\right] - \left[\frac{-2\pi^2 mK^2Z^2e^4}{n_1^2h^2}\right]$$

or, $$\frac{hc}{\lambda} = \frac{2\pi^2 mK^2Z^2e^4}{h^2}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\frac{1}{\lambda} = \frac{2\pi^2 mK^2e^4Z^2}{ch^3}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

where $\frac{2\pi^2 mK^2e^4}{ch^3}$ is a constant called **Rydberg constant (R)**

So, $$\bar{\nu} = \frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

* Value of R = 109677 cm^{-1} = 10967700m^{-1} $\simeq 1.09 \times 10^7$ m^{-1}

* $\frac{1}{R} = 912 Å$

Ex.31 What is the wavelength of light emitted when the electron in a hydrogen atom undergoes transition from the energy level with n = 4 to the energy level with n = 1 ?

Sol. $\bar{\nu} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

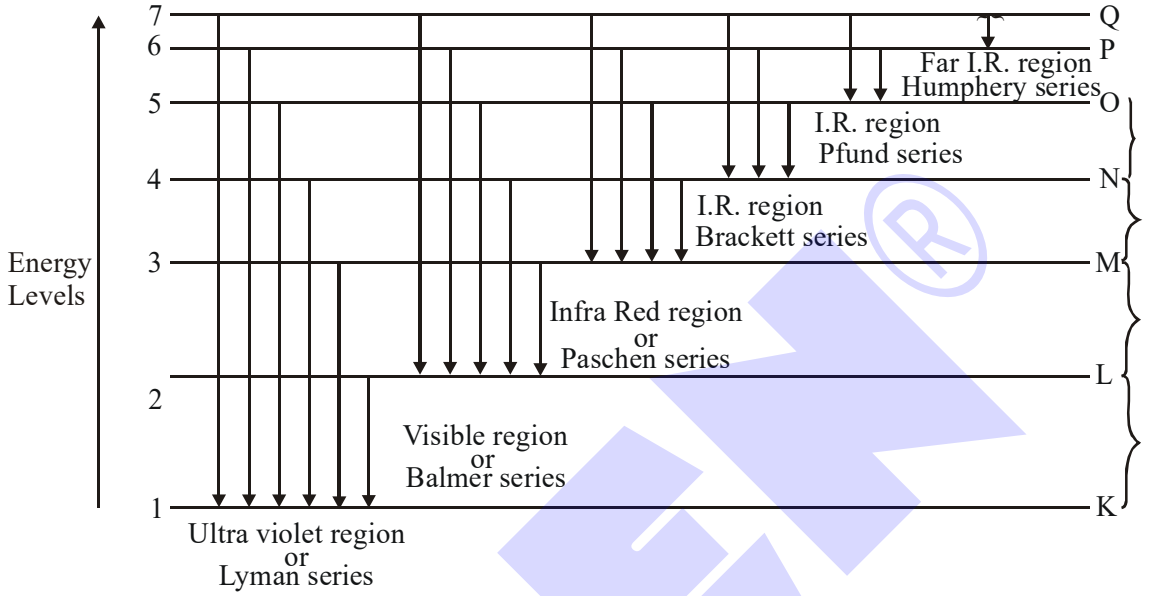
or, $$\frac{1}{\lambda} = R \times 1^2\left(\frac{1}{1^2} - \frac{1}{4^2}\right) = \frac{15R}{16}$$

$\therefore$ $\lambda = \frac{16}{15} \times \frac{1}{R} = 9.7859 \times 10^{-8}$ m = 978.59 Å.

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8.3 HYDROGEN LINE SPECTRUM :

When an electric excitation is applied on atomic hydrogen gas at low pressure,a bluish light is emitted. When a ray of this light is passed through a prism, a spectrum of several isolated sharp lines is obtained. The wavelength of various lines show that spectrum lines lie in visible, Ultraviolet and Infra red region. These lines are grouped into different series.



Series	Discovered by	Regions	n_2	n_1
Lyman	Lyman	U.V. region	$n_2 = 2,3,4 ...$	$n_1=1$
Balmer	Balmer	Visible region	$n_2 = 3,4,5 ...$	$n_1=2$
Paschen	Paschen	Infra red (I.R.)	$n_2 = 4,5,6 ...$	$n_1=3$
Brackett	Brackett	I.R. region	$n_2 = 5,6,7 ...$	$n_1=4$
Pfund	Pfund	I.R. region	$n_2 = 6,7,8 ...$	$n_1=5$
Humphery	Humphery	Far I.R. region	$n_2 = 7,8,9 ...$	$n_1=6$

8.3.1 KEY POINTS :

- First line / Starting line / Initial line ($\lambda_{max.}$ and ν_{min})
- Last line / limiting line / Series limit (λ_{min} and $\nu_{max.}$)
- First line of any series = α line
  Second line of any series = β line
  Third line of any series = γ line
- Total no. of emission lines between n_2 & $n_1 = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$, $(n_2 > n_1)$
- For transition from any orbit 'n' to n = 1 , total no. of emission lines $= \frac{n(n-1)}{2}$

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Ex.32 In a hydrogen spectrum if electron moves from orbit 6th to 2nd orbit by transition in multi steps, then find out the number of lines in spectrum

Sol. *Total number of line* $= 4 + 3 + 2 + 1 + 0 = 10$

Total number of lines $= \frac{(n_2 - n_1)\left[(n_2 - n_1)+1\right]}{2} = \frac{(6-2)(4+1)}{2} = 10$

Ex.33 In the spectrum of He^+ ion the wavelength of α line of Balamer series is x Å. What is the wavelength of β line of Paschen series.

Sol. $\frac{1}{\lambda_1} = Rz^2\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$

$\frac{1}{\lambda_2} = Rz^2\left(\frac{1}{3^2} - \frac{1}{5^2}\right)$

$\frac{\lambda_2}{\lambda_1} = \frac{\frac{1}{2^2} - \frac{1}{3^2}}{\frac{1}{3^2} - \frac{1}{5^2}} \quad \Rightarrow \quad \frac{\lambda_2}{x\text{Å}} = \frac{5}{16} \times \frac{25}{4}, \ \lambda_2 = 1.95 \text{ x (Å)}$

Ex.34 A sample of He^+ ions in ground state absorbs the radiation of x Å. subsequently, the sample emit radiation of 6 different wavelength. Calculate the value of x. $\left(\textbf{Take } \frac{1}{R} = 912 \text{ Å}\right)$

Sol. $\frac{n(n-1)}{2} = 6 \Rightarrow n = 4$

$\frac{1}{x} = R\left(\frac{1}{1^2} - \frac{1}{4^2}\right) \times 2^2$

$x = \frac{16}{15} \times \frac{912}{4} \text{Å} = 243.20$

Ex.35 Calculate the wavelength of the radiation of maximum frequency in the Lyman series of hydrogen spectrum.

Sol. For maximum ν, $n = \infty$ to $n = 1$.

Now, $\frac{1}{\lambda} = R \times 1^2\left(\frac{1}{1^2} - \frac{1}{\infty}\right)$

$\therefore \lambda = 912$ Å

8.4 Limitation of the Bohr's model

(i) Bohr's theory does not explain the spectrum of multi electron atom.

(ii) Why the Angular momentum of the revolving electron is equal to $\frac{nh}{2\pi}$, has not been explained by Bohr's theory.

(iii) Bohr's theory does not explain the fine structure of spectral lines. Fine structure of the spectral line is obtained when spectrum is viewed by a spectroscope of high resolving power.

(iv) Bohr's theory does not explain the splitting of spectral lines in the presence of magnetic field **(Zeeman effect)** or electric field **(Stark effect)**

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9. DUAL BEHAVIOUR OF MATTER & DE BROGLIE WAVELENGTH :

In 1924, a French physicist, ***Louis de Broglie*** suggested that, like light, matter also has dual character. It exhibits wave as well as particle nature. According to de Broglie, the wavelength λ of any moving particle is inversely proportional to its momentum p.



de Broglie'
(1892-1987)

A French physicist, studied history as an undergraduate in the early 1910,s. His interest turned to science as a result of his assignment to radio communications in world war 1. He was awarded the Nobel Prize in physics in 1929.

$$\lambda \propto \frac{1}{p}$$

$$\boxed{\lambda = \frac{h}{p} = \frac{h}{mv}}$$

Here h = Planck's constant

p = momentum of particle

It is clear from the above equation that the value of λ decreases on increasing either m or v or both. The wavelength of many fast-moving objects like an aeroplane or a cricket ball, is very low because of their high mass. Thus wave nature of macroscopic objects can be neglected but for microscopic particles like electrons, protons, atoms etc. wave nature is significant & cannot be neglected.

9.1 DERIVATION OF BOHR'S ANGULAR MOMENTUM QUANTIZATION RULE :

We know that according to Bohr theory, $mvr = \frac{nh}{2\pi}$

or $2\pi r = \frac{nh}{mv}$

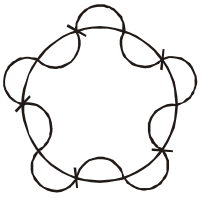
or $2\pi r = \frac{nh}{p} = n\lambda$ $\left(\because \frac{h}{p} = \lambda \text{ de-Broglie equation}\right)$

According to de Broglie, an electron bound to the nucleus behaves like a standing wave.
A standing wave, also known as a stationary wave, is a wave that remains in a constant position.

For a circular standing wave to persist, a whole number of wavelength must fit into the circumference of the circle (2π).
And if n number of waves of λ wavelength are present in this circle total circumference will be nλ.

$2\pi r = n\lambda$

n = Number of wave made by electron in one complete revolution.



n = 5

∴ **Waves made = 5**

Electron in n^{th} orbit of any unielectron system can create 'n' waves in one complete revolution.

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Ex.36 Calculate the de Broglie wavelength of a ball of mass 0.1 kg moving with a speed of 30 ms^{-1}.

Sol. $\lambda = \frac{h}{mv} = \frac{6.6\times10^{-34}}{0.1\times30} = 2.2 \times 10^{-34}\ m$

This is apparent that this wavelength is too small for ordinary observation.

Although the de Broglie equation is applicable to all material objects but it has significance only in case of microscopic particles.

Ex.37 What should be the mass of the sodium photon if its wavelength is 5894 Å. The speed of light is 3×10^8 metre/second and the value of h is 6.652×10^{-34} kg m^2/sec.?

(A) 3.746×10^{-26} ***(B)*** 3.746×10^{-30}

(C) 3.746×10^{-34} ***(D)*** 3.746×10^{-36}

Sol. $\lambda = \frac{h}{m\times c} \Rightarrow m = \frac{h}{c\lambda}$

$(\because \lambda = 5894Å = 5894 \times 10^{-10}\ m)$

$m = \frac{6.652\times10^{-34}}{3\times10^{-8}\times5894\times10^{-10}}$ *or* $\frac{6.652}{17682}\times10^{-32}$

$= 0.0003746 \times 10^{-32} = 3.746 \times 10^{-36}\ kg$

Ex.38 Calculate the de-Broglie wavelength of electron accelerated by the following potential differences.

(i) 750 V (ii) 300 volt

Sol. *(i)* $\lambda = \sqrt{\frac{150}{V}}Å = \sqrt{\frac{150}{750}}Å = \frac{1}{\sqrt{5}}Å$ *(ii)* $\lambda = \sqrt{\frac{150}{300}}Å = \frac{1}{\sqrt{2}}Å$

Ex.39 Find de-Broglie wavelength of electron with KE = 9.6×10^{-19} J.

Sol. $KE = \frac{9.6\times10^{-19}}{1.6\times10^{-19}}eV = 6eV$

$\lambda = \sqrt{\frac{150}{6}}Å = 5Å$

KE of 6 eV means e^- is accelerated by 6 volt.

Ex.40 Calculate the ratio of de-Broglie wavelength of electron and α-particle.

(i) Moving at same speed

(ii) Moving at same momentum

(iii) Having same K.E.

(iv) Accelerated from rest through the same P.D.

Sol. *(i)* $\lambda = \frac{n}{mv}$

$\lambda \propto \frac{1}{m}$

$\frac{\lambda_{electron}}{\lambda_\alpha} = \frac{m_\alpha}{m_e} = \frac{4\times1836}{1}$

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(ii) $\frac{\lambda_{electron}}{\lambda_\alpha} = \frac{1}{1}$

(iii) $\frac{h}{\sqrt{2mE}}$

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{\lambda_e}{\lambda_\alpha} = \sqrt{\frac{m_\alpha}{m_e}} = \sqrt{\frac{1836\times4}{1}}$$

(iv) $\lambda = \frac{h}{\sqrt{2mqV}} = \frac{\lambda_e}{\lambda_\alpha} = \sqrt{\frac{(mq)_\alpha}{(mq)_e}} = \sqrt{\frac{4\times2}{1/1836\times1}}$

Ex.41 In Li^{2+} ion electron jumps from 2nd to 1st orbit. If the emitted radiation is absorbed by H atom. Calculate the de-Broglie wavelength of the ejected electron.

Sol. $\Delta E = 13.6\times9\left(1-\frac{1}{4}\right) = 91.8eV$

Excess energy = 91.8 – 13.6 = 78.2 eV

$$\lambda = \sqrt{\frac{150}{78.2}}\text{Å} = 1.38\text{Å}$$

Ex.42 Photoelectrons are liberated by ultra violet light of wavelength 2000 Å from a metallic surface for which the photoelectric threshold is 4000 Å. Calculate the de-Broglie wavelength of electrons emitted with maximum kinetic energy.

Solution : *K.E. = Quantum Energy – Threshold energy*

$$= \frac{6.626\times10^{-34}\times3\times10^8}{2000\times10^{-10}} - \frac{6.626\times10^{-34}\times3\times10^8}{4000\times10^{-10}}$$

$$= \frac{6.626\times10^{-34}\times3\times10^8}{10^{-10}}\left(\frac{1}{2000}-\frac{1}{4000}\right) = 4.969\times10^{-19}\ \text{Joule.}$$

$\frac{1}{2}mv^2 = 4.969\times10^{-19} \Rightarrow m^2v^2 = 2\times4.969\times10^{-19}\times9.1\times10^{-31}$

$mv = 9.51\times10^{-25} \Rightarrow \lambda = \frac{h}{mv} = \frac{6.626\times10^{-34}}{9.51\times10^{-25}} = 0.696\times10^{-9}\ m$

Ex.43 Calculate the de-Broglie wavelength when proton is accelerated by the 750 V.

Sol. $\lambda = \sqrt{\frac{150}{V\times1836}}\text{Å} = \sqrt{\frac{150}{750\times1836}}\text{Å} = 0.01\ \text{Å}.$

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9.2 Justification of dual nature of electrons :

I. Particle character :

(a) If an electron strikes a screen coated with ZnS , it produces a spot of light called scintillation, one electron produces only one scintillation point which means electron are localised not spread out like wave.

(b) Electron possess definite mass, momentum & KE proving their particle nature.

II. Wave character :

- It was confirmed by phenomenon of diffraction, interference, reflection.

  Davisson and Germer showed that when high speed electron strike Ni crystal a diffraction pattern (having number of rings) is obtained like X-rays of electromagnetic spectrum.

10. HEISENBERG UNCERTAINTY PRINCIPLE :

Bohr's theory considers an electron as a material particle. Its position and momentum can be determined with accuracy. But, when an electron is considered in the form of wave as suggested by de-Broglie, it is not possible to ascertain simultaneously the exact position and speed of the electron more precisely at a given instant since the wave is extending throughout a region of space.

In ***1927, Werner Heisenberg*** presented a principle known as Heisenberg uncertainty principle which states as : ***"It is impossible to measure simultaneously the exact position and exact momentum of a body as small as an electron".***

The uncertainty of measurement of position, Δx, and the uncertainty of momentum Δp or $m\Delta v$, are related by Heisenberg's relationship as : ($p = mv$, $\Delta p = m\Delta v$)

$$\Delta x \,.\, \Delta p \geq \frac{h}{4\pi} \quad \text{or} \quad \Delta x \,.\, m\Delta v \geq \frac{h}{4\pi}$$

or $$\Delta \mathbf{x}.\Delta \mathbf{v} \geq \frac{\mathbf{h}}{\mathbf{4\pi m}}$$

where h is Planck's constant.

In the case of bigger particles (having considerable mass), the value of uncertainty product is negligible.

Ex.44 A golf ball has a mass of 40 g and a speed of 45 m/s. If the speed can be measured within accuracy of 2 % , calculate the uncertainty in the position.

Sol. Mass of the ball $= 40\ g = 40 \times 10^{-3}\ kg$

The uncertainty in the speed,

$$\Delta v = 45 \times \frac{2}{100} = 0.9\,ms^{-1}$$

$$\Delta x = \frac{h}{4\pi m\,\Delta v} = \frac{6.626\times10^{-34}\,Js}{4\times3.14\times(40\times10^{-3}\,kg)(0.9\,ms^{-1})} = 1.46 \times 10^{-33}\ m.$$

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Ex.45 Calculate the uncertainty in the velocity of a cricket ball of mass 150 g, if the uncertainty in its position in of the order of 1 Å.

Sol. Mass of ball ,

$m = 150\ g = 150 \times 10^{-3}\ kg = 0.150\ kg$

Uncertainty in position, $\Delta x = 1\ Å = 10^{-10}\ m$

$$\Delta x \times m\Delta v = \frac{h}{4\pi}$$

$$\Delta v = \frac{h}{4\pi \times \Delta x \times m}$$

$$= \frac{6.626 \times 10^{-34}\ kg\, m^2\, s^{-1}}{4 \times 3.14 \times 10^{-10}\ m \times 0.150\ kg}$$

$= 3.52 \times 10^{-24}\ m\ s^{-1}$.

11. QUANTUM MECHANICAL MODEL OF ATOM

Classical mechanics, based on Newton's laws of motion, successfully describes the motion of all macroscopic objects such as a falling stone, orbiting planets etc., which have essentially a particle-like behaviour as shown in the previous section. However it fails when applied to microscopic objects like electrons, atoms, molecules etc. This is mainly because of the fact that classical mechanics ignores the concept of dual behaviour of matter especially for sub-atomic particles and the uncertainty principle. The branch of science that takes into account this dual behaviour of matter is called **quantum mechanics.**

Quantum mechanics is a theoretical science that deals with the study of the motions of the microscopic objects that have both observable wave like and particle like properties. It specifies the laws of motion that these objects obey. When quantum mechanics is applied to macroscopic objects (for which wave like properties are insignificant) the results are the same as those from the classical mechanics.

Quantum mechanics was developed independently in 1926 by Werner Heisenberg and Erwin Schrodinger. Here, however, we shall be discussing the quantum mechanics which is based on the ideas of wave motion. The fundamental equation of quantum mechanics was developed by **Schrodinger** and it won him the Nobel Prize in Physics in 1933. This equation which incorporates wave-particle duality of matter as proposed by de Broglie is quite complex and knowledge of higher mathematics is needed to solve it. You will learn its solutions for different systems in higher classes.

For a system (such as an atom or a molecule whose energy does not change with time) the Schrodinger equation is written as

$$\hat{H}\psi = E\psi$$

$\hat{H}$ is a mathematical operator called **Hamiltonian**. Schrodinger gave a recipe of constructing this operator from the expression for the total energy of the system. The total energy of the system takes

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into account the kinetic energies of all the sub-atomic particles (electrons, nuclei), attractive potential between the electrons and nuclei and repulsive potential among the electrons and nuclei individually. Solution of this equation gives E and ψ.

11.1 *HYDROGEN ATOM AND THE SCHRODINGER EQUATION*

When Schrodinger equation is solved for hydrogen atom, the solution gives the possible energy levels the electron can occupy and the corresponding wave function(s) (ψ) of the electron associated with each energy level. These quantized energy states and corresponding wave functions which are characterized by a set of three quantum numbers (**principal quantum number *n*, azimuthal quantum number *l* and magnetic quantum number** m_l) arise as a natural consequence in the solution of the Schrodinger equation. When an electron is in any energy state, the wave function corresponding to that energy state contains all information about the electron.



Erwin Schrodinger (1887-1961)

Erwin Schrodinger, an Austrian physicist received his Ph.D. in theoretical physics from the University of Vienna in 1910. In 1927 Schrodinger succeeded Max Planck at the University of Berlin at Planck's request. In 1933, Schrodinger left Berlin because of his opposition to Hitler and Nazi policies and returned to Austria in 1936. After the invasion of Austria by Germany, Schrodinger was forcibly removed from his professorship. He then moved to Dublin, Ireland where he remained for seventeen years. Schrodinger shared the Nobel Prize for Physics with P.A.M. Dirac in 1933.

The wave function is a mathematical function whose value depends upon the coordinates of the electron in the atom and does not carry any physical meaning. Such wave functions of hydrogen or hydrogen like species with one electron are called **atomic orbitals**.

Such wave functions pertaining to one-electron species are called one-electron systems. **The probability of finding an electron at a point within an atom is proportional to the $|\psi|^2$ at that point.** The quantum mechanical results of the hydrogen atom successfully predict all aspects of the hydrogen atom spectrum including some phenomena that could not be explained by the Bohr model.

Application of Schrodinger equation to multi-electron atoms presents a difficulty: the Schrodinger equation cannot be solved exactly for a multi-electron atom. This difficulty can be overcome by using approximate methods. Such calculations with the aid of modern computers show that orbitals in atoms other than hydrogen do not differ in any radical way from the hydrogen orbitals discussed above. The principal difference lies in the consequence of increased nuclear charge. Because of this all the orbitals are somewhat contracted. Further, as you shall see later , unlike orbitals of hydrogen or hydrogen like species, whose energies depend only on the quantum number n, the energies of the orbitals in multi-electron atoms depend on quantum numbers *n* and *l*.

11.2 Important Features of the Quantum Mechanical Model of Atom–

Quantum mechanical model of atom is the picture of the structure of the atom, which emerges from the application of the Schrodinger equation to atoms. The following are the important features of the quantummechanical model of atom:

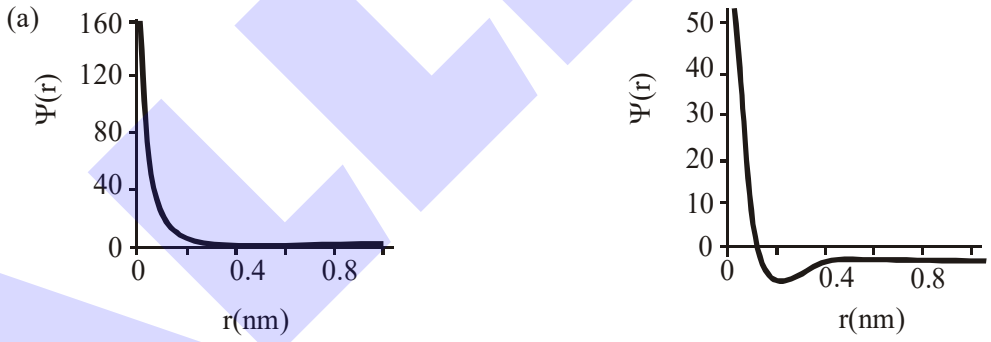
1. The energy of electrons in atoms is quantized (i.e., can only have certain specific values), for example when electrons are bound to the nucleus in atoms.
2. The existence of quantized electronic energy levels is a direct result of the wave like properties of electrons and are allowed solutions of Schrodinger wave equation.

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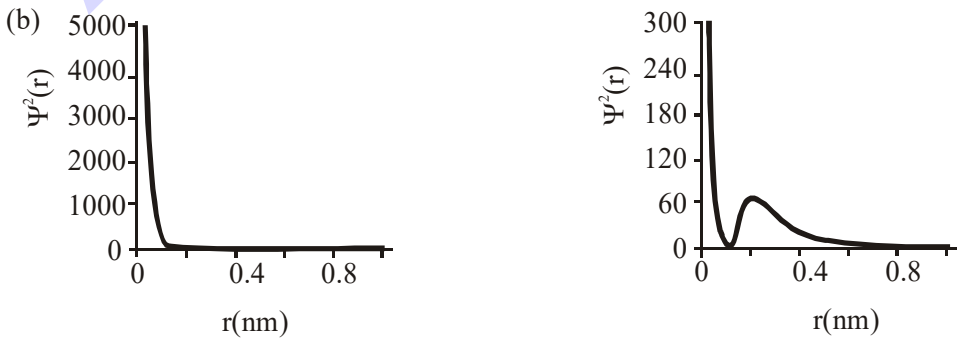
3. Both the exact position and exact velocity of an electron in an atom cannot be determined simultaneously (Heisenberg uncertainty principle). The path of an electron in an atom therefore, can never be determined or known accurately. That is why, as you shall see later on, one talks of only probability of finding the electron at different points in an atom.
4. **An atomic orbital is the wave function ψ for an electron in an atom.** Whenever an electron is described by a wave function, we say that the electron occupies that orbital. Since many such wave functions are possible for an electron, there are many atomic orbitals in an atom. These "one electron orbital wave functions" or orbitals form the basis of the electronic structure of atoms. In each orbital, the electron has a definite energy. An orbital cannot contain more than two electrons. In a multi-electron atom, the electrons are filled in various orbitals in the order of increasing energy. For each electron of a multi-electron atom, there shall, therefore, be an orbital wave function characteristic of the orbital it occupies. All the information about the electron in an atom is stored in its orbital wave function ψ and quantum mechanics makes it possible to extract this information out of ψ.
5. The probability of finding an electron at a point within an atom is proportional to the square of the orbital wave function i.e., $|\psi|^2$ at that point. $|\psi|^2$ is known as **probability density** and is always positive. From the value of $|\psi|^2$ at different points within an atom, it is possible to predict the region around the nucleus where electron will most probably be found.

11.3 Shapes of Atomic Orbitals

The orbital wave function or ψ for an electron in an atom has no physical meaning. It is simply a mathematical function of the coordinates of the electron. However, for different orbitals the plots of corresponding wave functions as a function of r (the distance from the nucleus) are different. Such plots for 1s (n = 1, l = 0) and 2s (n = 2, l = 0) orbitals are

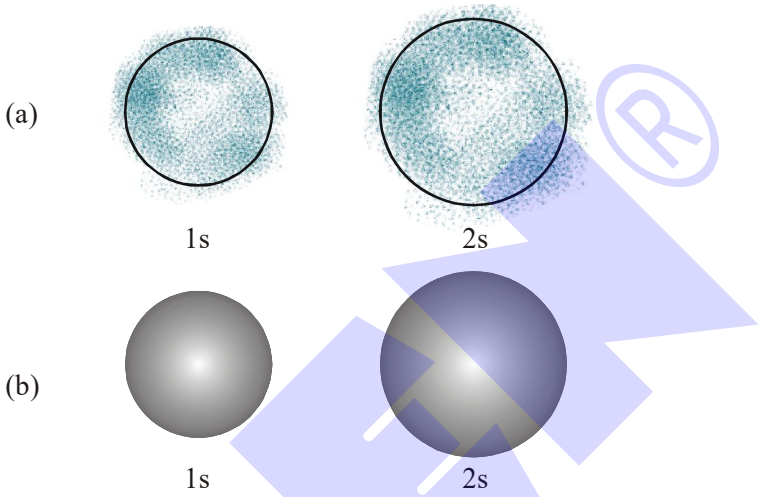


According to the German physicist, Max Born, the square of the wave function (i.e., ψ^2) at a point gives the probability density of the electron at that point. The variation of ψ^2 as a function of r for 1s and 2s orbitals is given in fig. Here again, you may note that the curves for 1s and 2s orbitals are different.



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It may be noted that for 1s orbital the probability density is maximum at the nucleus and it decreases sharply as we move away from it. On the other hand, for 2s orbital the probability density first decreases sharply to zero and again starts increasing. After reaching a small maxima it decreases again and approaches zero as the value of r increases further. The region where this probability density function reduces to zero is called **nodal surfaces** or simply **nodes**. In general, it has been found that ns-orbital has (n – 1) nodes, that is, number of nodes increases with increase of principal quantum number n. In other words, number of nodes for 2s orbital is one, two for 3s and so on.
These probability density variation can be visualised in terms of charge cloud diagrams. In these diagrams, the density of the dots in a region represents electron probability density in that region.

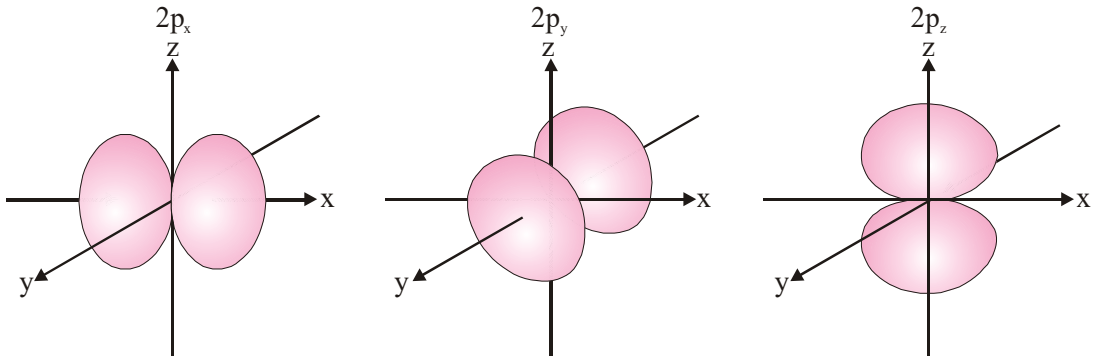


(a) Probability density plots of 1s and 2s atomic orbitals. The density of the dots represents the probability density of finding the electron in that region. (b) Boundary surface diagram for 1s and 2s orbitals.
Boundary surface diagrams of constant probability density for different orbitals give a fairly good representation of the shapes of the orbitals. In this representation, a boundary surface or contour surface is drawn in space for an orbital on which the value of probability density $|\psi|^2$ is constant. In principle many such boundary surfaces may be possible. However, for a given orbital, only that boundary surface diagram of constant probability density is taken to be good representation of the shape of the orbital which encloses a region or volume in which the probability of finding the electron is very high, say, 90%. The boundary surface diagram for 1s and 2s orbitals are given in fig. One may ask a question : Why do we not draw a boundary surface diagram, which bounds a region in which the probability of finding the electron is, 100 %? The answer to this question is that the probability density $|\psi|^2$ has always some value, howsoever small it may be, at any finite distance from the nucleus. It is therefore, not possible to draw a boundary surface diagram of a rigid size in which the probability of finding the electron is 100%. Boundary surface diagram for a s orbital is actually a sphere centred on the nucleus. In two dimensions, this sphere looks like a circle. It encloses a region in which probability of finding the electron is about 90%.

Thus we see that 1s and 2s orbitals are spherical in shape. In reality all the s-orbitals are spherically symmetric, that is, **the probability of finding the electron at a given distance is equal in all the directions**. It is also observed that **the size of the s orbital increases with increase in n, that is, 4s > 3s > 2s > 1s** and the electron is located further away from the nucleus as the principal quantum number increases.

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Boundary surface diagrams for three 2p orbitals (l = 1) are

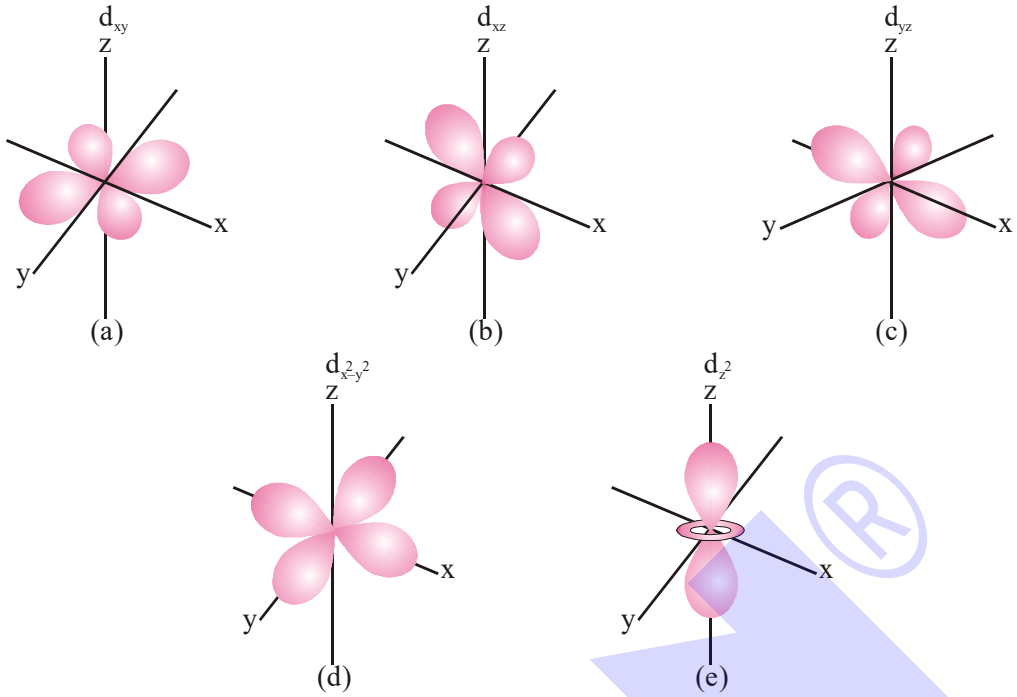


Boundary surface diagrams of the three 2p orbitals.

In these diagrams, the nucleus is at the origin. Here, unlike s-orbitals, the boundary surface diagrams are not spherical. Instead each p orbital consists of two sections called lobes that are on either side of the plane that passes through the nucleus. The probability density function is zero on the plane where the two lobes touch each other. The size, shape and energy of the three orbitals are identical. They differ however, in the way the lobes are oriented. Since the lobes may be considered to lie along the x, y or z axis, they are given the designations $2p_x$, $2p_y$, and $2p_z$. It should be understood, however, that there is no simple relation between the values of m_l (–1, 0 and +1) and the x, y and z directions. For our purpose, it is sufficient to remember that, because there are three possible values of m_l, there are, therefore, three p orbitals whose axes are mutually perpendicular. Like s orbitals, **p orbitals increase in size and energy with increase in the principal quantum number** and hence the order of the energy and size of various p orbitals is **4p > 3p > 2p**. Further, like s orbitals, the probability density functions for p-orbital also pass through value zero, besides at zero and infinite distance, as the distance from the nucleus increases. The number of nodes are given by the n – 2, that is number of radial node is one for 3p orbital, two for 4p orbital and so on.

For $l = 2$, the orbital is known as d-orbital and the minimum value of principal quantum number (n) has to be 3. as the value of l cannot be greater than n – 1. There are five m_l values –2, –1, 0, +1 and +2) for $l = 2$ and thus there are five d orbitals. The boundary surface diagram of d orbitals are :

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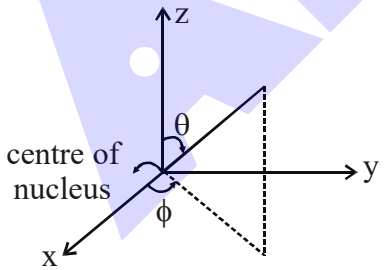
Boundary surface diagrams of the five 3d orbitals.

The five d-orbitals are designated as d_{xy}, d_{yz}, d_{xz}, $d_{x^2-y^2}$ and d_{z^2}. The shapes of the first four d-orbitals are similar to each other, where as that of the fifth one, d_{z^2}, is different from others, but all five 3d orbitals are equivalent in energy. The d orbitals for which n is greater than 3 (4d, 5d...) also have shapes similar to 3d orbital, but differ in energy and size.

Besides the radial nodes (i.e., probability density function is zero), the probability density functions for the np and nd orbitals are zero at the plane (s), passing through the nucleus (origin). For example, in case of p_z orbital, xy-plane is a nodal plane, in case of d_{xy} orbital, there are two nodal planes passing through the origin and bisecting the xy plane containing z-axis. These are called **angular nodes** and number of angular nodes are given by '*l*', i.e., one angular node for p orbitals, two angular nodes for 'd' orbitals and so on.

The total number of nodes are given by (n–1), i.e., sum of l angular nodes and (n – l – 1) radial nodes.

11.4 SOLUTION OF SCHRODINGER EQUATION :



Spherical coordinate system

The solution in spherical coordinates may be represented as :

$\psi = R(r).\Theta(\theta).\Phi(\phi)$

R(r) = Radial function depends on n and l

$\Theta(\theta).\Phi(\phi)$ = Angular function depends on l and m = $A(\theta, \phi) = Y(\theta, \phi)$.

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11.4.1 Radical part of solution :

1s $(n = 1, \ell = 0)$: $R_{1s}(r) = 2\cdot\left(\frac{z}{a_0}\right)^{3/2}\cdot e^{-\sigma/2}$

where $\sigma = \frac{2Zr}{na_0}$ $\quad a_0$ = I[st] Bohr's radius = 0.529Å

2s $(n = 2, \ell = 0)$: $R_{2s}(r) = \frac{1}{2\sqrt{2}}\cdot\left(\frac{Z}{a_0}\right)^{3/2}\cdot(2-\sigma)e^{-\sigma/2}$

2p $(n = 2, \ell = 1)$: $R_{2p}(r) = \frac{1}{2\sqrt{6}}\cdot\left(\frac{Z}{a_0}\right)^{3/2}\cdot\sigma\cdot e^{-\sigma/2}$

3s $(n = 3, \ell = 0)$: $R_{3s}(r) = \frac{1}{9\sqrt{3}}\cdot\left(\frac{Z}{a_0}\right)^{3/2}\cdot(6-6\sigma+\sigma^2)e^{-\sigma/2}$

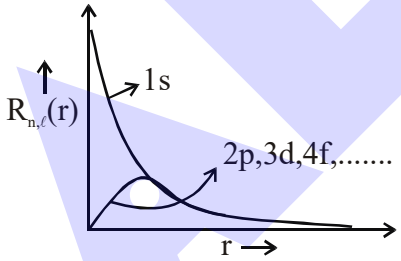
3p $(n = 3, \ell = 1)$: $R_{3p}(r) = \frac{1}{9\sqrt{6}}\cdot\left(\frac{Z}{a_0}\right)^{3/2}\cdot\sigma(4-\sigma)e^{-\sigma/2}$

3d $(n = 3, \ell = 2)$: $R_{3d}(r) = \frac{1}{9\sqrt{30}}\cdot\left(\frac{Z}{a_0}\right)^{3/2}\cdot\sigma^2\cdot e^{-\sigma/2}$

General form : $R_{n\ell}(r) = K.e^{-\sigma/2}.\sigma^{\ell}$ (Polynomial of order $n - \ell - 1$)

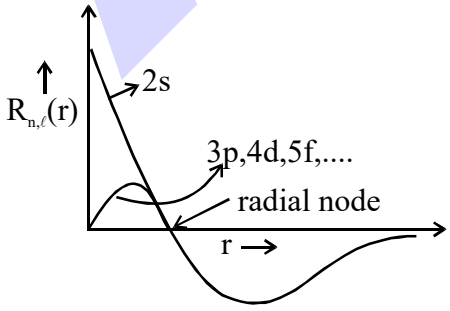
11.4.2 Graph of radial function [R(r) or Ψ(r)] :

Only the graph of s-orbital does not start from origin.

(i) 

$n - \ell - 1 = 0$

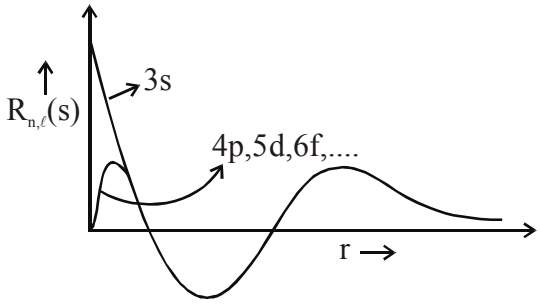
1s, 2p, 3d, 4f .....orbitals

(ii) 

$n - \ell - 1 = 1$

2s, 3p, 4d, 5f .....orbitals

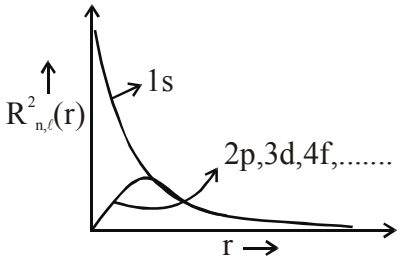
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(iii) 

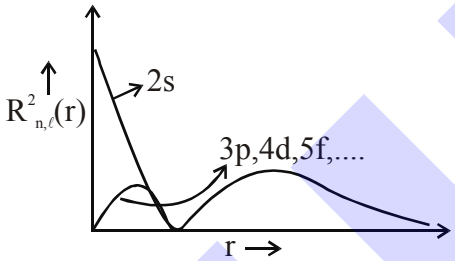
$n - \ell - 1 = 2$

3s, 5p, 5d, 6f ..... orbitals

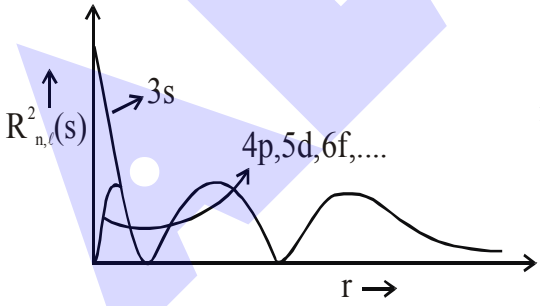
11.4.3 GRAPH OF RADIAL PROBABLITY DENSITY FUNCTION $[R^2(r)$ or $\psi^2(r)]$:

(i) 

$n - \ell - 1 = 0$, (1s, 2p, 3d, 4f .....orbitals)

(ii) 

$n - \ell - 1 = 1$ (2s, 3p, 4d, 5f .....orbitals)

(iii) 

$n - \ell - 1 = 2$ (3s, 5p, 5d, 6f ..... orbitals)

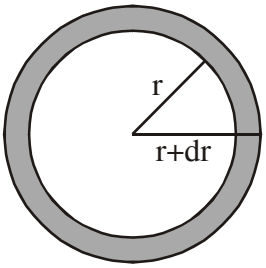
11.4.4 RADIAL PROBABILITY DISTRIBUTION FUNCTION (RPDF), $4\pi r^2\psi^2(r)$

It is often useful to know the likelihood of finding the electron in an orbital at any given distance away from the nucleus. This enables us to say at what distance from the nucleus the electron is most likely to be found, and also how tightly or loosely the electron is bound in a particular atom. This is expressed by the radial probability distribution function, $4\pi r^2\psi^2(r)$.

Radial distribution function is the measure of the probability of finding the electron in a spherical shell between thickness r and (r + dr) from the nucleus, irrespective of the direction.

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❖ **Volume of radial shell :**



$$dV = \begin{bmatrix}\text{Volume of sphere}\\ \text{with radius}(r+dr)\end{bmatrix} - \begin{bmatrix}\text{Volume of sphere}\\ \text{with radius } r\end{bmatrix}$$

$$= \frac{4}{3}\pi(r+dr)^3 - \frac{4}{3}\pi r^3$$

$$= \frac{4}{3}\pi(r^3 + 3r^2dr + 3rdr^2 + dr^3) - \frac{4}{3}\pi r^3 = \frac{4}{3}\pi[r^3 + 3r^2dr - r^3]$$

(As dr represents an extremely small thickness, the higher powers of dr such as dr^2 and dr^3 may be neglected.)

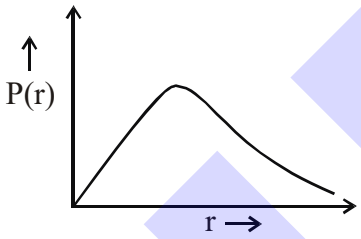
$\therefore$ Volume of shell, $dV = \frac{4}{3}(\pi \times 3r^2dr) = 4\pi r^2dr$

Now, radial probability density, $R^2(r) = \frac{P}{dV}$

$\therefore$ Probability of finding electron in the volume element,

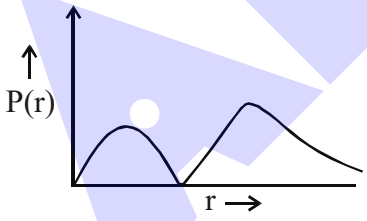
$P = R^2(r).\ dV = R^2(r).4\pi r^2.dr$

Now radial probability distribution function , $P(r) = \frac{P}{dr} = 4\pi r^2.R^2(r)$

(i) 

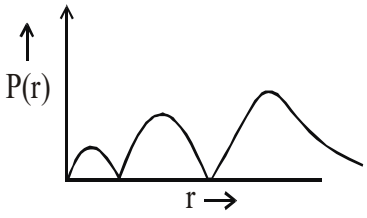
$n - \ell - 1 = 0$

1s, 2p, 3d, 4f .....orbitals

(ii) 

$n - \ell - 1 = 1$

2s, 3p, 4d, 5f .....orbitals

(iii) 

$n - \ell - 1 = 2$

3s, 5p, 5d, 6f ..... orbitals

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❖ **Chracteristics of radial distribution function :**

(i) The number of maxima in radial distribution function plot are $(n - \ell)$.

(ii) The maximum probability of finding the electron, for the ground state hydrogen atom (1s) is found to be at a_0 (first Bohr radius).

(iii) For 2s, 3s, 3p orbitals , the number of maxima is more than one, indicating that there is maximum probability of finding the electron at the distance corresponding to the highest value of peak.

However, there is lesser probability of finding the electron at the other peaks. It shows that in a certain state, the electron spends some portion of its time very close to the nucleus.

11.5 ANGULAR PART OF SOLUTION :

(i) s-orbital :

$\ell = 0,\ m = 0$ $\qquad \Theta(\theta)\ .\ \Phi(\phi) = \sqrt{\frac{1}{4\pi}}$

As the probability of finding electron is not depending on angle (direction) then it must be same. In all direction and hence the shape of s-orbital is sphere.

(ii) p-orbital :

p_x -orbital : $\ell = 1,\ m = \pm 1$ $\qquad \Theta(\theta)\ .\ \Phi(\phi) = \sqrt{\frac{3}{4\pi}} \cdot \sin\theta\cos\phi$

p_y -orbital : $\ell = 1,\ m = \pm 1$ $\qquad \Theta(\theta)\ .\ \Phi(\phi) = \sqrt{\frac{3}{4\pi}} \cdot \sin\theta\sin\phi$

p_z -orbital : $\ell = 1,\ m = 0$ $\qquad \Theta(\theta)\ .\ \Phi(\phi) = \sqrt{\frac{3}{4\pi}} \cdot \cos\theta$

(iii) d-orbital :

d_{z^2} -orbital : $\ell = 2,\ m = 0$ $\qquad \Theta(\theta)\ .\ \Phi(\phi) = \sqrt{\frac{5}{16\pi}}(3\cos^2\theta - 1)$

$d_{x^2-y^2}$ -orbital : $\ell = 2,\ m = \pm 2$ $\qquad \Theta(\theta)\ .\ \Phi(\phi) = \sqrt{\frac{15}{4\pi}} \cdot \sin^2\theta\cos 2\phi$

d_{xy} -orbital : $\ell = 2,\ m = \pm 2$ $\qquad \Theta(\theta)\ .\ \Phi(\phi) = \sqrt{\frac{15}{4\pi}} \cdot \sin^2\theta\sin 2\phi$

d_{xz} -orbital : $\ell = 2,\ m = \pm 1$ $\qquad \Theta(\theta)\ .\ \Phi(\phi) = \sqrt{\frac{15}{4\pi}} \cdot \sin\theta\cos\theta\cos\phi$

d_{yz} -orbital : $\ell = 2,\ m = \pm 1$ $\qquad \Theta(\theta)\ .\ \Phi(\phi) = \sqrt{\frac{15}{4\pi}} \cdot \sin\theta\cos\theta\cdot\sin\phi$

Note : (i) Number of radial nodes = $n - \ell - 1$

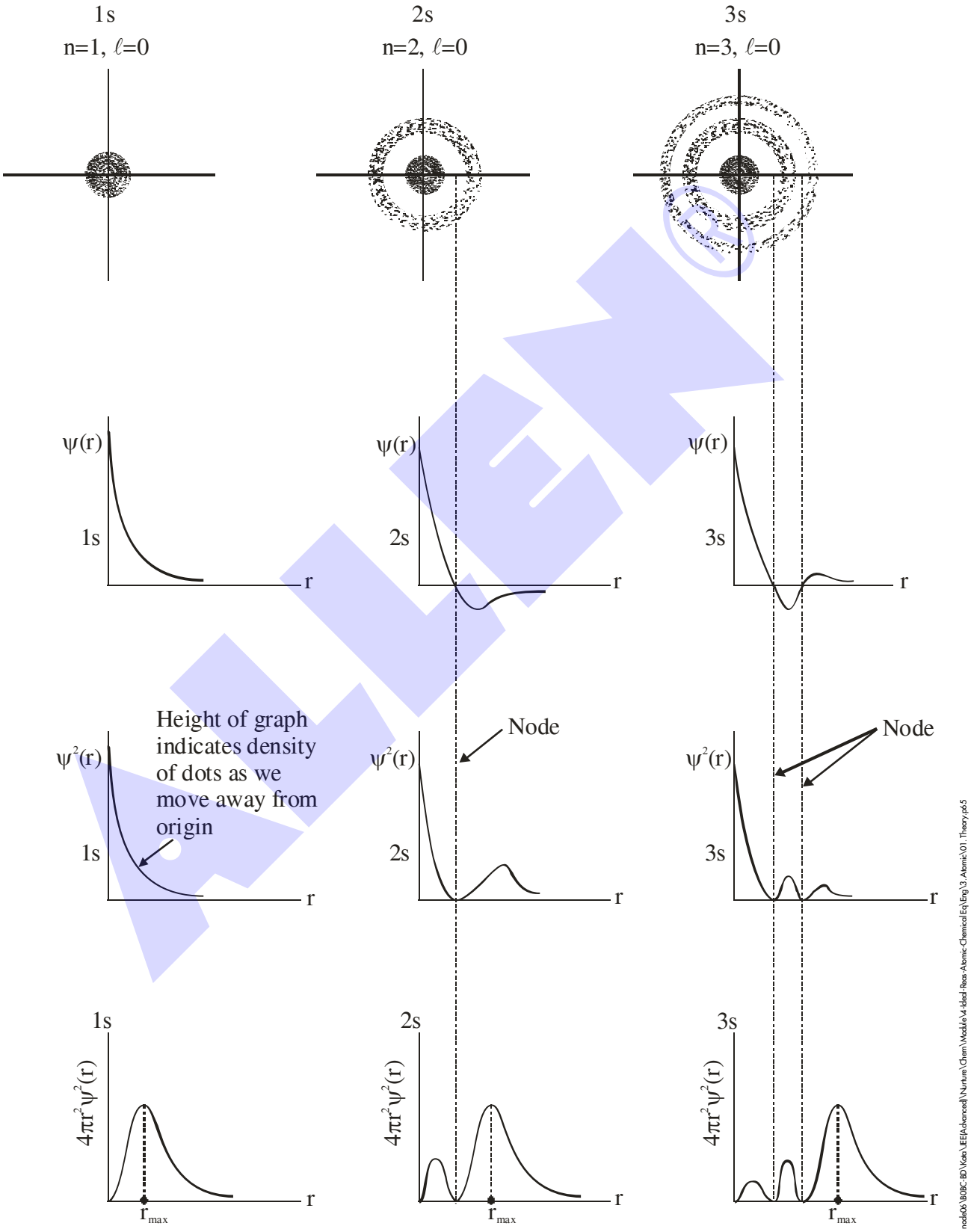
Number of angular nodes = ℓ

Total of number nodes = $n - 1$

(ii) On standard axis consideration, the orbital with m = 0 have angular dependency only on θ and they represent orbitals with z-component like p_z, d_{z^2} , etc.

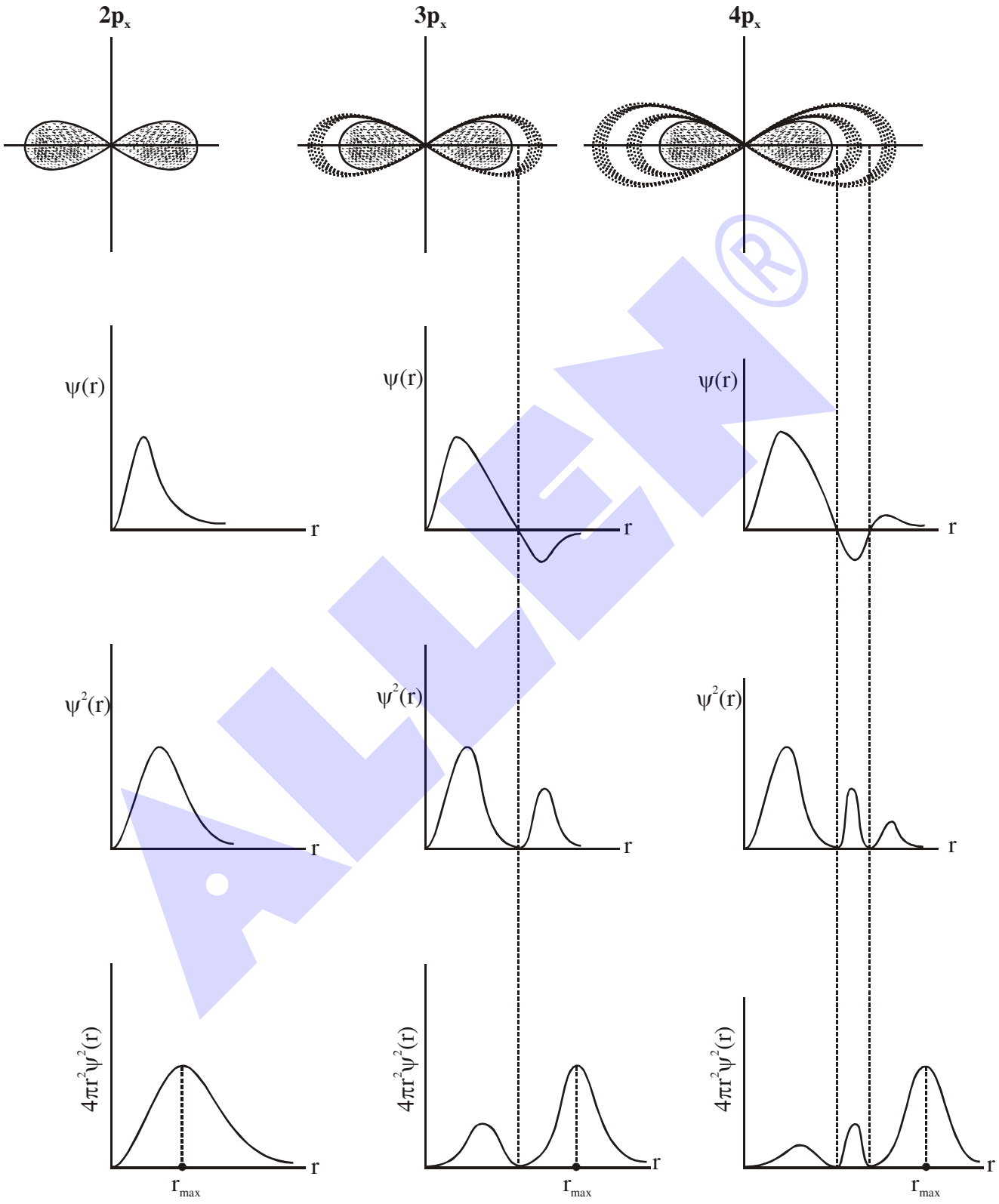
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11.6 Electron-density distribution in 1s, 2s and 3s-orbitals. The lower part of the fig. shows how the electron density, represented by ψ^2 varies as a function of distance from the nucleus. In the 2s and 3s - orbitals, the electron-density function drops to zero at certain distances from the nucleus. The spherical surfaces around the nucleus at which ψ^2 is zero are called nodes.



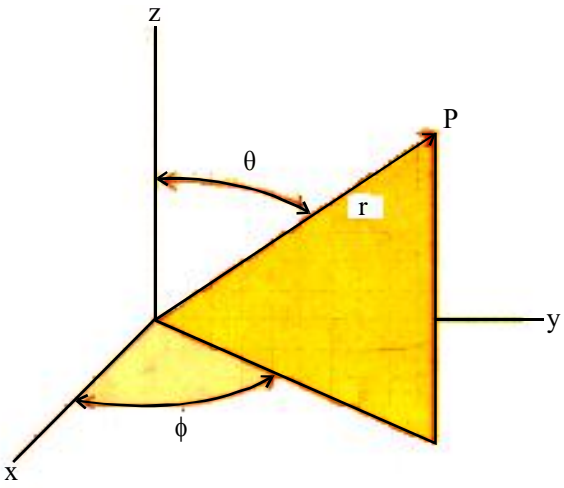
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11.7 Electron-density distribution in $2p_x$, $3p_x$ and $4p_x$-orbitals :

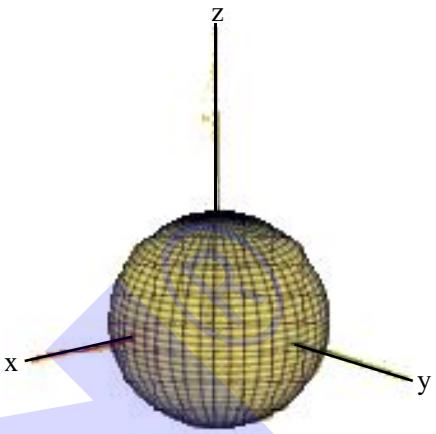


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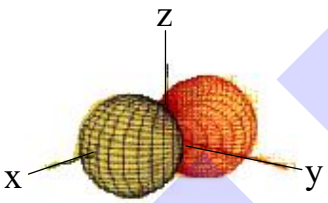
11.8 SHAPES OF ATOMIC ORBITALS



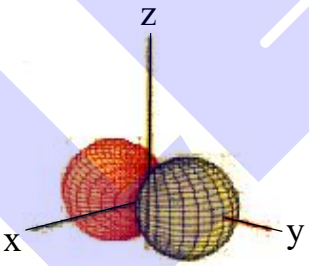
The spherical Polar Coordinates



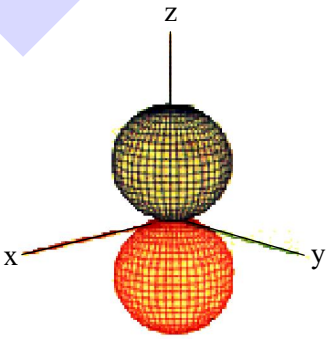
S



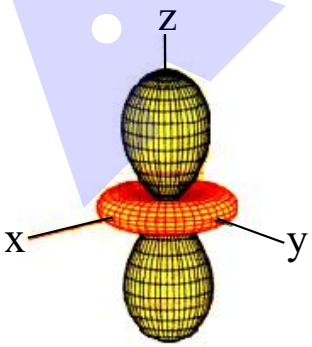
p_X



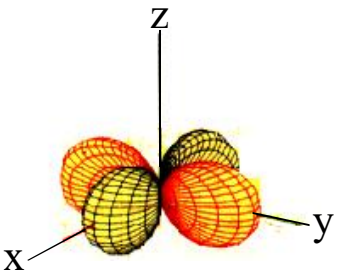
p_y



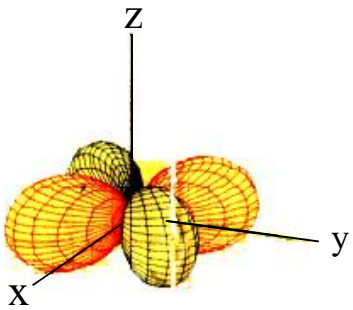
p_z



d_{z^2}

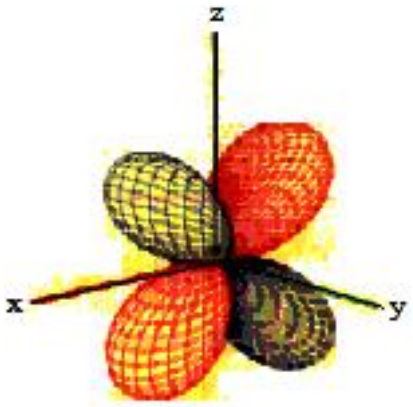


$d_{x^2-y^2}$

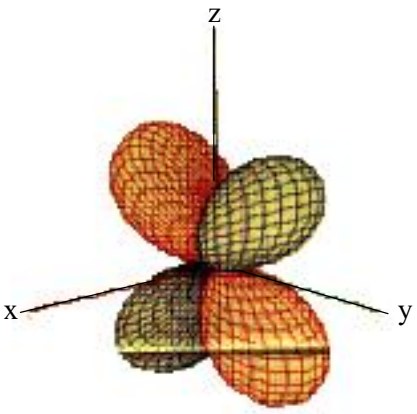


d_{xy}

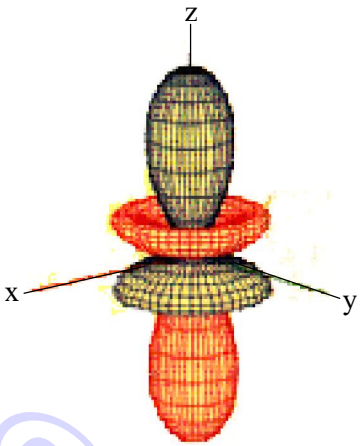
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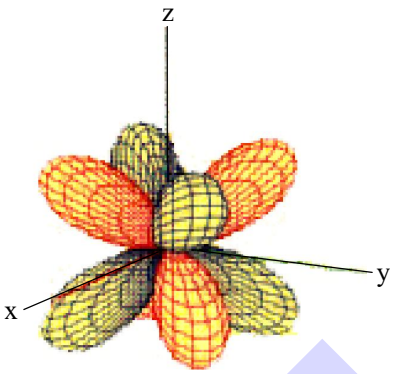
d_{xz}



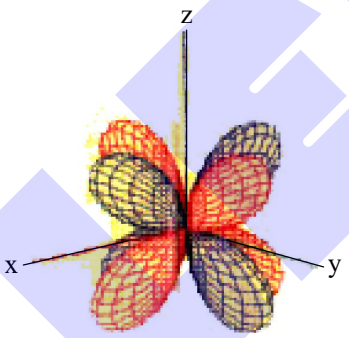
d_{yz}



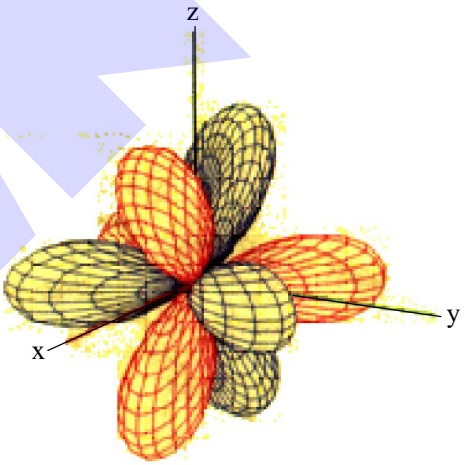
f_{z^3}



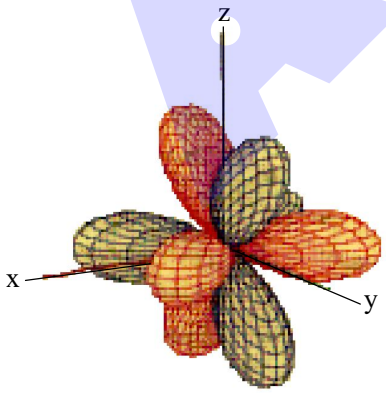
f_{xyz}



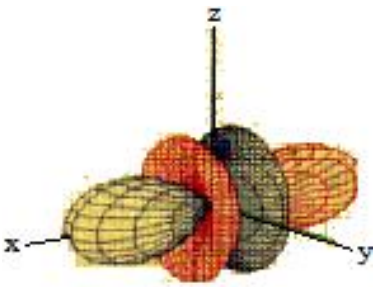
$f_{z(x^2-y^2)}$



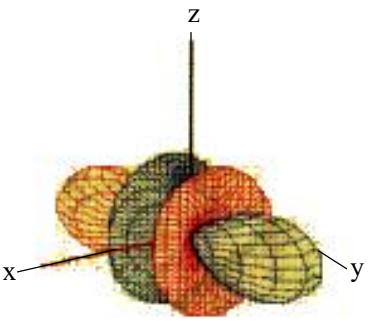
$f_{x(y^2-z^2)}$



$f_{y(z^2-x^2)}$



f_{x^3}



f_{y^3}

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Ex.46 Find the distance at which probability of finding electron is maximum for 1s orbital in a hydrogen atom. The wave function of orbital is given as.

$$\psi_{1s} = \frac{2}{a_0^{3/2}}.e^{-\frac{r}{a_0}}$$

Sol. Probability distribution function is $P(r) = \psi^2.4\pi r^2 = \left(\frac{4}{a_0^3}\right)e^{\left(-\frac{2r}{a_0}\right)}.4\pi r^2$

or, $P(r) = k \cdot r^2.e^{\frac{-2r}{a_0}}$

Now, $\frac{dP(r)}{dr} = \frac{16\pi}{a_0^3}\left[2r\left(e^{-\frac{2r}{a_0}}\right) - r^2 e^{-\frac{2r}{a_0}} \times \left(-\frac{2}{a_0}\right)\right] = 0$

or, $\frac{16\pi}{a_0}.2r.e^{\frac{-2r}{a_0}}\left(1-\frac{r}{a_0}\right) = 0$

or $r = a_0$

$\Rightarrow$ probability of finding electron is maximum at distance a_0 from nucleus.

Ex.47 Consider ψ (wave function) of 2s atomic orbital of H-atom is -

$$\psi_{2s} = \frac{1}{4\sqrt{2\pi}\,a_0^{3/2}}\left[2-\frac{r}{a_0}\right]e^{-\frac{r}{2a_0}}$$

Find distance of radial node from nucleus in terms of a_0.

Sol. $R(r) = 0$

$$\left[2-\frac{r}{a_0}\right]e^{-\frac{r}{2a_0}} = 0 \Rightarrow 2-\frac{r}{a_0} = 0 \Rightarrow r = 2\,a_0$$

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SELF STUDY

12. SUB-ATOMIC PARTICLES

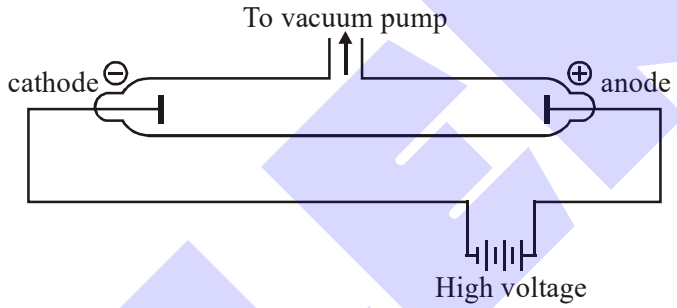
Dalton's atomic theory was able to explain the law of conservation of mass, law of constant composition and law of multiple proportion very successfully. However, it failed to explain the results of many experiments, for example, it was known that substances like glass or ebonite when rubbed with silk or fur generate electricity. Many different kinds of sub-atomic particles were discovered in the twentieth century.

12.1 Discovery of Electron :

In 1830, Michael Faraday showed that if electricity is passed through a solution of an electrolyte, chemical reactions occurred at the electrodes, which resulted in the liberation and deposition of matter at the electrodes. These results suggested the particulate nature of electricity.

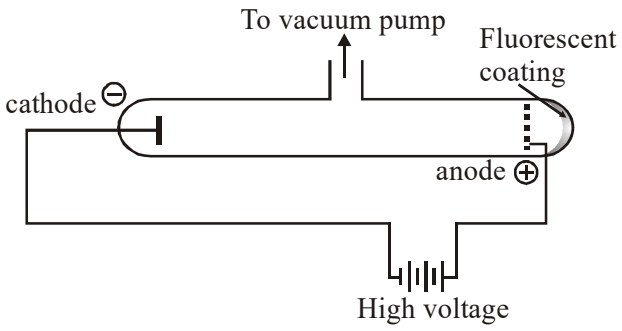
An insight into the structure of atom was obtained from the experiments on electrical discharge through gases. Before we discuss these results we need to keep in mind a basic rule regarding the behaviour of charged particles: "Like charges repel each other and unlike charges attract each other".

In mid 1850s many scientists mainly Faraday began to study electrical discharge in partially evacuated tubes, known as **cathode ray discharge tubes**.



A cathode ray discharge tube

A cathode ray tube is made of glass containing two thin pieces of metal, called electrodes, sealed in it. The electrical discharge through the gases could be observed only at very low pressures and at very high voltages. The pressure of different gases could be adjusted by evacuation. When sufficiently high voltage is applied across the electrodes, current starts flowing through a stream of particles moving in the tube from the negative electrode (cathode) to the positive electrode (anode). These were called **cathode rays or cathode ray particles**. The flow of current from cathode to anode was further checked by making a hole in the anode and coating the tube behind anode with phosphorescent material zinc sulphide. When these rays, after passing through anode, strike the zinc sulphide coating, a bright spot on the coating is developed(same thing happens in a television set)



A cathode ray discharge tube with perforated anode

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The results of these experiments are summarised below.

(i) The cathode rays start from cathode and move towards the anode.

(ii) These rays themselves are not visible but their behaviour can be observed with the help of certain kind of materials (fluorescent or phosphorescent) which glow when hit by them. Television picture tubes are cathode ray tubes and television pictures result due to fluorescence on the television screen coated with certain fluorescent or phosphorescent materials.

(iii) In the absence of electrical or magnetic field, these rays travel in straight lines.

(iv) In the presence of electrical or magnetic field, the behaviour of cathode rays are similar to that expected from negatively charged particles, suggesting that the cathode rays consist of negatively charged particles, called **electrons**.

(v) The characteristics of cathode rays (electrons) do not depend upon the material of electrodes and the nature of the gas present in the cathode ray tube.

Thus, we can conclude that electrons are basic constituent of all the atoms.

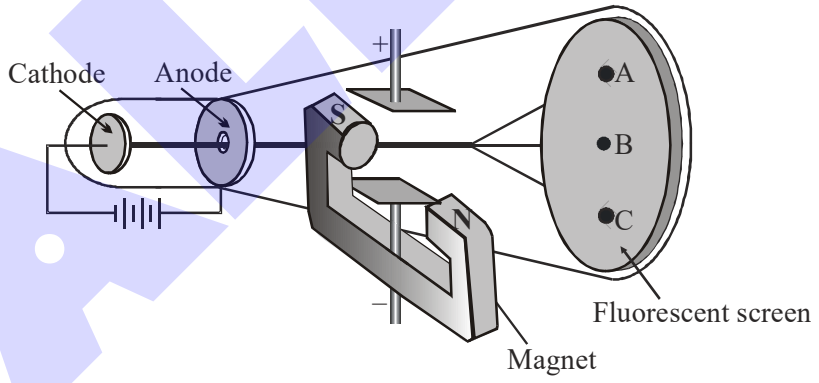
12.1.1 Charge to Mass Ratio of Electron :

In 1897, British physicist J.J. Thomson measured the ratio of electrical charge (e) to the mass of electron (m_e) by using cathode ray tube and applying electrical and magnetic field perpendicular to each other as well as to the path of electrons. Thomson argued that the amount of deviation of the particles from their path in the presence of electrical or magnetic field depends upon :

(i) the magnitude of the negative charge on the particle, greater the magnitude of the charge on the particle, greater is the interaction with the electric or magnetic field and thus greater is the deflection.

(ii) the mass of the particle lighter the particle, greater the deflection.

(iii) the strength of the electrical or magnetic field . the deflection of electrons from its original path increases with the increase in the voltage across the electrodes, or the strength of the magnetic field.



The apparatus to determine the charge to the mass ratio of electron

When only electric field is applied, the electrons deviate from their path and hit the cathode ray tube at point A. Similarly when only magnetic field is applied, electron strikes the cathode ray tube at point C. By carefully balancing the electrical and magnetic field strength, it is possible to bring back the electron to the path followed as in the absence of electric or magnetic field and they hit the screen at point B. By carrying out accurate measurements on the amount of deflections observed by the electrons on the electric field strength or magnetic field strength, Thomson was able to determine the value of e/me as:

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$$\frac{e}{m_e} = 1.758820 \times 10^{11} C\,kg^{-1}$$

where m_e is the mass of the electron in kg and e is the magnitude of the charge on the electron in coulomb (C). Since electrons are negatively charged, the charge on electron is –e.

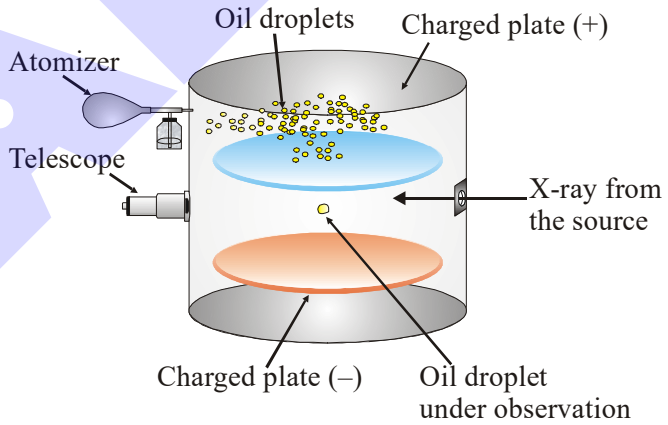
12.1.2 Charge on the Electron :

R.A. Millikan (1868-1953) devised a method known as oil drop experiment (1906-14), to determine the charge on the electrons. He found that the charge on the electron to be -1.6×10^{-19} C. The present accepted value of electrical charge is -1.6022×10^{-19} C. The mass of the electron (m_e) was determined by combining these results with Thomson's value of e/m ratio.

$$m_e = \frac{e}{e/m_e} = \frac{1.6022 \times 10^{-19} C}{1.758820 \times 10^{11} C\,kg^{-1}} = 9.1094 \times 10^{-31} kg$$

Millikan's Oil Drop Method

In this method, oil droplets in the form of mist, produced by the atomiser, were allowed to enter through a tiny hole in the upper plate of electrical condenser. The downward motion of these droplets was viewed through the telescope, equipped with a micrometer eye piece. By measuring the rate of fall of these droplets, Millikan was able to measure the mass of oil droplets.The air inside the chamber was ionized by passing a beam of X-rays through it. The electrical charge on these oil droplets was acquired by collisions with gaseous ions. The fall of these charged oil droplets can be retarded, accelerated or made stationary depending upon the charge on the droplets and the polarity and strength of the voltage applied to the plate. By carefully measuring the effects of electrical field strength on the motion of oil droplets, Millikan concluded that the magnitude of electrical charge, q, on the droplets is always an integral multiple of the electrical charge, e, that is, q = n e, where n = 1, 2, 3... .
The highest common factor (HCF) of charges on oil droplet was taken as charge on electron.

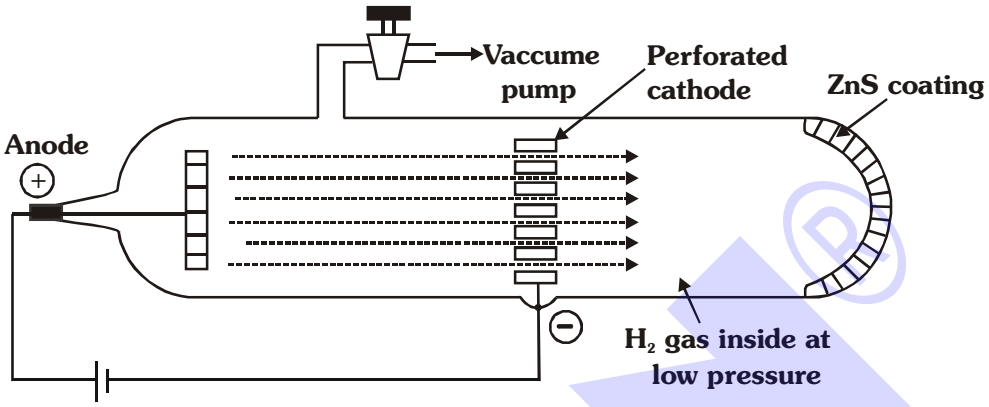


The Millikan oil drop apparatus for measuring charge 'e'. In chamber, the forces acting on oil drop are: gravitational, electrostatic due to electrical field and a viscous drag force when the oil drop is moving.

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12.2 CANAL RAYS (OR ANODE RAYS) – DISCOVERY OF PROTON :

Atoms are electrically neutral. Hence after the discovery of the negatively charged constituent (electron) of an atom, attempts were made to discover the positively charged counterpart of electrons. By using a discharge tube containing a perforated cathode, ***Goldstein (1886)*** found that some rays passed through these holes in a direction opposite to that of the cathode rays.



Positive Rays or Canal Rays

Electrical discharge carried out in the modified cathode ray tube led to the discovery of particles carrying positive charge, also known as **canal rays**. The characteristics of these positively charged particles are listed below.

(i) unlike cathode rays, the positively charged particles depend upon the nature of gas present in the cathode ray tube. These are simply the positively charged gaseous ions.

(ii) The charge to mass ratio of the particles is found to depend on the gas from which these originate.

(iii) Some of the positively charged particles carry a multiple of the fundamental unit of electrical charge.

(iv) The behaviour of these particles in the magnetic or electrical field is opposite to that observed for electron or cathode rays.

The smallest and lightest positive ion was obtained from hydrogen and was called proton. This positively charged particle was characterised in 1919.

The lightest positively charged particle is called a proton (P or P+). Positive rays are atomic or molecular cations from which some electrons have been removed. The removed electrons constitute the cathode rays and the positive cations form the positive or canal rays.

12.3 DISCOVERY OF NEUTRON :

After the discovery of electron and proton ***Rutherford (1920)*** predicted the existence of a neutral fundamental particle. In ***1932, Chadwick*** bombarded the element Beryllium with α-particles and noticed the emission of a radiation having the following characteristics.

(i) The radiation was highly penetrating.

(ii) The radiation was unaffected by magnetic and electric fields which shows that it is electrically neutral.

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(iii) It was found to have approximately the same mass as the protons.

The name 'neutron' was given to this sub-atomic particle. It is denoted by n or ${}_0n^1$. Bombardment of beryllium by α-particles results in the formation of carbon and neutrons are emitted.

$${}_4Be^9 + {}_2He^4 \longrightarrow {}_6C^{12} + {}_0n^1$$

Mass of a neutron is 1.00867 amu (1.67493×10^{-24}g or 1.67493×10^{-27} kg)

12.4 PROPERTIES OF FUNDAMENTAL PARTICLES :

Atoms are made up-essentially, of three fundamental particles, which differ in mass and electric charge as follows :

	Electron	**Proton**	**Neutron**
Symbol	e or e^-	p	n
Approximate relative mass	1/1836	1	1
Approximate relative charge	–1	+1	0
Mass in kg	9.10982×10^{-31}	$1.6726216 \times 10^{-27}$	1.674927×10^{-27}
Mass in amu	0.00054	1.00727	1.00867
Actual charge/C	$-1.602176 \times 10^{-19}$	$+1.602176 \times 10^{-19}$	0

Ex.48 Arrange the particle in their increasing order of specific charge ratio.

(a) e^-, P, n, α-particle ***(b) Na^+, Li^+, F^-, Mg^{2+}, Al^{3+}***

Sol. ***(a)*** $\left(\frac{e}{m}\right)_e = \frac{1e}{\left(\frac{1}{1836}\right)amu}$

$$\left(\frac{e}{m}\right)_P = \frac{1}{1} = 1 = \frac{1e}{1amu}$$

$$\left(\frac{e}{m}\right)_n = \frac{0}{1} = 0 = \frac{0e}{1amu}$$

$$\left(\frac{e}{m}\right)_\alpha = \frac{2}{4} = \frac{1}{2} = \frac{2e}{4amu}$$

Ans. $n < \alpha < p < e^-$

(b) $Na^+ = \frac{1}{23}$; $Li^+ = \frac{1}{7}$; $F^- = \frac{1}{19}$; $Mg^{2+} = \frac{2}{24} = \frac{1}{12}$; $Al^{3+} = \frac{3}{27} = \frac{1}{9}$

$Na^+ < F^- < Mg^{2+} < Al^{3+} < Li^+$

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Ex.49 Which of the following pairs have same specific charge $\left(\frac{e}{m}\right)$ ***?***

(a) electron & proton ***(b) electron & positron*** ***(c) proton & positron***
(d) proton & deutron ***(e) α-particle & deutron***

Answer. b, e

Sol. Positron is antiparticle of electron (same charge and mass, but positively charged).

Ex.50 Through what potential difference an α-particle should be accelerated to have speed 5×10^6 m/s.

Sol. $qV = \frac{1}{2}mv^2$

$$\left(2\times1.6\times10^{-19}\times V\right) = \frac{1}{2}\times\left(4\times1.66\times10^{-27}\right)\times\left(5\times10^{6}\right)^2$$

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SOME PREVIOUS YEARS QUESTION

1. ***With what velocity should an α–particle travel towards the nucleus of a Cu atom so as to arrive at a distance 10^{-13} m .*** **[JEE 1997]**

Ans. **6.3×10^6 m/s**

Sol. $\frac{1}{2}mv^2 = \frac{kq_1q_2}{r}$

$v^2 = \frac{2kq_1q_2}{mr}$

$= \frac{2 \times 9 \times 10^9 \times 2e \times 29e}{(10^{-13})\ (1.67 \times 10^{-27})}$

$= \frac{18 \times 2 \times 29 \times 10^9 \times (1.6)^2 \times 10^{-38}}{1.67 \times 10^{-23} \times 10^{-27}}$

$= \frac{36 \times 29 \times 2.56 \times 10^{-29+40}}{1.67}$

$= 1600.3 \times 10^{11}$

2. ***A compound of Vanadium has magnetic moment of 1.73 BM work out electronic configuration of Vanadium Ion in the compound.*** **[JEE 1997]**

Ans. **[Ar] $3d^1$**

Sol. $r = \sqrt{n(n+2)}$ $n = 1$

$= \sqrt{3}$

$= 1.73$ [Ar] $3d^2$

3. ***The energy of an electron in the first Bohr orbit of H atom is –13.6 eV. The possible energy value(s) of the excited state(s) for electrons in Bohr orbits of hydrogen is/are :***

(A) – 3.4 eV (B) – 4.2 eV (C) – 6.8 eV (D) + 6.8 eV **[JEE 1998]**

Ans. **(A)**

Sol. $E = -13.6$

$E = \frac{-13.6}{4} = -3.4$ eV

4. ***The number of nodal planes in a p_x orbital is:*** **[JEE 2000]**

(A) one (B) two (C) three (D) zero

Ans. **(A)**

Sol. **One**

5. ***Calculate the energy required to excite one litre of hydrogen gas at 1 atmp and 298K to the first excited state of atomic hydrogen. The energy for the dissociation of H – H is 436 KJ mol^{-1}.***

Ans. **97.819 KJ**

Sol. $E = 436$

$T = 298$ K

$p = 1$ atm Energy req. $= 436 \times \frac{1}{298 \times R}$

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$v = 1\ell$ $\quad E = \left(10.2 \times 96.47 \times \frac{2}{R \times 298}\right) + \left(\frac{436}{R \times 298}\right)$

$n = \frac{1}{298R} = \frac{2 \times 04 \times 96.47 \times 436}{298 \times R}$

6. The quantum numbers +1/2 and –1/2 for the electron spin represent: **[JEE 2001]**

(A) rotation of the electron in clockwise and anticlockwise direction respectively.

(B) rotation of the electron in anticlockwise and clockwise direction respectively.

(C) magnetic moment of the electron pointing up and down respectively.

(D) two quantum mechanical spin states which have no classical analogue.

Ans. **(D)**

Sol. Two quantum mechanical spin states which have no classical analogue.

7. Rutherfords experiment , which established the nuclear model of atom, used a beam of :–

(A) β - particles, which impinged on a metal foil and get absorbed. **[JEE 2002]**

(B) γ - rays, which impinged on a metal foil and ejected electron.

(C) Helium atoms, which impinged on a metal foil and got scattered.

(D) Helium nuclie, which impinged on a metal foil and got scattered.

Ans. **(D)**

Sol. Helium nuclie, which impinged on a metal foil and got scattered.

8. The magnetic moment of cobalt of the compund $Hg[Co(SCN)_4]$ is [Given : Co^{+2}]

(A) $\sqrt{3}$ (B) $\sqrt{8}$ (C) $\sqrt{15}$ (D) $\sqrt{24}$ **[JEE 2004]**

Ans. **(C)**

Sol. Co^{+2}

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^7$

$n = 3$

$r = \sqrt{n(n+2)}$

$= \sqrt{3 \times 5} = \sqrt{15}$

9. The radius of which of the following orbit is same as that of the first Bohr's orbit of hydrogen atom?

(A) He^+ $(n = 2)$ (B) Li^{2+} $(n = 2)$ (C) Li^{2+} $(n = 3)$ (D) Be^{3+} $(n = 2)$ **[JEE 2004]**

Ans. **(D)**

Sol. $r = 0.529 \times \frac{n^2}{z}$ $\quad r_H = 0.529$

$r_{He^+} = 0.529 \times \frac{4}{2}$ $\quad r_{Li^{+2}} = 0.529 \times \frac{9}{3}$

$r_{Li^{+2}} = 0.529 \times \frac{4}{3}$ $\quad r_{Be^{+3}} = 0.529 \times \frac{4}{4} = r_H$

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10. *(a) The Schrodinger wave equation for hydrogen atom is* **[IIT-2004]**

$$\psi_{2s} = \frac{1}{4(2\pi)^{1/2}}\left(\frac{1}{a_0}\right)^{3/2}\left(2-\frac{r_0}{a_0}\right)e^{-r/a}$$

Where a_0 is Bohr's radius. Let the rdial node in 2s be at r_0. Then find r_0 in terms of a_0.

(b) A base ball having mass 100 g moves with velocity 100 m/s. Find out the value of wavelength of base ball.

Ans. **(a) $r_0 = 2a_0$, (b) 6.626×10^{-25} Å**

Sol. $\Psi_{2s} = \frac{1}{4(2\pi)^{1/2}}\left(\frac{1}{a_0}\right)^{3/2}\left(2-\frac{r_0}{a_0}\right)e^{-r/a}$

(b) $\lambda = \frac{h}{mv}$

$= \frac{6.626\times10^{-34}}{(0.1)(100)}$

$= 6.626 \times 10^{-35}$ m

$= 6.626 \times 10^{-35} \times 10^{+10}$

$= 6.626 \times 10^{-25}$ Å

(a) $2-\frac{r_0}{a_0}=0$

$\frac{r_0}{a_0} = 2$

$r_0 = 2a_0$

11. *(a) Calculate velocity of electron in first Bohr orbit of hydrogen atom (Given $r = a_0$)* **[IIT-2005]**

(b) Find de-Broglie wavelength of the electron in first Bohr orbit.

(c) Find the orbital angular momentum of 2p-orbital in terms of $h/2\pi$ units.

Ans. **(a) 2.197×10^6 m/s (b) 3.31 Å (c) $\sqrt{2}.\frac{h}{2\pi}$**

Sol. (a) $v = 2.188\times10^6 \times \frac{1}{1}\text{m/s}$

$= \frac{6.626\times10^{-34}}{9.1\times10^{-31}\times2.188\times10^6}$

$= 0.332 \times 10^{-9}$ m

$= 3.31 \times 10^{-10}$ m

$= 3.31$ Å

(b) $\lambda = \frac{h}{mv}$

(c) $\sqrt{\ell(\ell+1)}\frac{h}{2\pi}$

$= \sqrt{1(1+1)}\frac{h}{2\pi} = \frac{\sqrt{2}h}{2\pi}$

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12. Given in hydrogenic atom r_n, V_n, E, K_n stand for radius, potential energy, total energy and kinetic energy in n^{th} orbit. Find the value of U,v,x,y. **[JEE 2006]**

(A)	$U = \frac{V_n}{K_n}$	(P)	1
(B)	$\frac{1}{r_n} \propto E^x$	(Q)	–2
(C)	$r_n \propto Z^y$ (Z = Atomic number)	(R)	–1
(D)	v = (Orbital angular momentum of electron in its lowest energy)	(S)	0

Ans. (A) - Q ; (B) - P ; (C) - R ; (D) - S

Sol. (A) $U = \frac{V_n}{K_n}$

$= \frac{-27.2}{13.6} = -2$

(B) $\frac{1}{r_n} \propto E^x$

(C) $r_n \propto Z^y$

$0.529\ n^2/z = (z)^y$

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13. Match the entries in Column I with the correctly related quantum number(s) in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS. **[JEE 2008]**

	Column I		**Column II**
(A)	Orbital angular momentum of the electron in a hydrogen-like atomic orbital	(P)	Principal quantum number
(B)	A hydrogen-like one-electron wave function obeying Pauli principle	(Q)	Azimuthal quantum number
(C)	Shape, size and orientation of hydrogen like atomic orbitals	(R)	Magnetic quantum number
(D)	Probability density of electron at the nucleus in hydrogen-like atom	(S)	Electron spin quantum number

Ans. **(A) - Q,R ; (B) - P,Q,R,S ; (C) - P,Q,R ; (D) - P,Q**

Sol. The direction of orbital angular momentum is given by magnetic quantum number.

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Important Notes

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